MILITARY SPECIFICATION

Weapons Delivery Computer for the Bombing Set, Angle Rate AN/ASB-19(V)

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers the requirements for a digital computer designated as weapons delivery computer (WDC) for the AN/ASB-19(V) Angle Rate Bombing Set (ARBS).

1.2 Classification. The equipment covered by this specification shall consist of the following items:

Type Designation	Applicable paragraph
Computer, Digital CP-1276/ASB-19(V)	3.5.1
Converter-Interface Unit, Computer, J-3429/ASB-19(V)	3.5.2

1.3 Associated Equipment. This equipment shall operate with the associated equipment listed in 6.6.

2. APPLICABLE DOCUMENTS

2.1 General. Unless otherwise specified herein, the following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall have precedence.

SPECIFICATIONS

Military

MIL-P-116	Presentation - Packaging, Methods
MIL-C-172	Cases; Bases, Mounting; and Mounts, Vibration (For use with Electronic Equipment in Aircraft)

Comments or recommendations which may be of use in improving this document may be sent to the preparing activity when a Self-Addressed Standardization Document Improvement Proposal (DD Form 1426) appears at the end of this document, or by letter addressed to the cognizant activity.

FSC 1280

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MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-N-7793	Meter, Time Totalizing
MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aircraft, Installation of
MIL-D-8706	Data and Test Engineering, Contract Requirements for Aircraft Weapons Systems
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-23103	Thermal Performance Evaluaation, Airborne Electronic Equipment; General Requirement for
MIL-M-81288	Mounting Bases, Flexible Plastic Foam
MIL-C-81511	Connector, Electrical, Circular, High Density, Quick Disconnect; Environment Resisting and Accessories, General Specification for
MIL-A-85055	Bombing Set, Angle Race AN/ASB-19(V)

Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of
AR-10	Maintainability of Avionics Equipment and Systems, General Requirements for

STANDARDS

Military

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-198	Capacitors, Selection and Use of
MIL-STD-411	Climate Extremes for Military Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment

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MIL-STD-461A	Electromagnetic Interference Characteristics, Requirements for Equipment, Notice 3, 1 May 1970
MIL-STD-462	Electromagnetic, Interference Characteristics, Measurement of, Notice 2, 1 May 1970
MIL-STD-471	Maintainability Demonstration
MIL-STD-480	Configuration Control - Engineering Changes, Deviations and Waivers
MIL-STD-681	Identification Coding and Application of Hook-Up and Lead Wire
MIL-STD-701	List of Standard Semiconductor Devices
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MXL-STD-781B	Reliability Tests, Exponential Distribution
MIL-STD-785	Reliability Program for System and Equipment Development and Production
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810B	Environmental Test Methods
MIL-STD-882	System Safety Program for Systems and Associated Subsystem and Equipment, Requirements for
MIL-sTD-1378	Standard Hardware Program Modules, Requirements for Employing
MIL-STD-1472	Human Engineering Design Criteria for Military Systems Equipment and Facilities
MIL-STD-2074(AS)	Failure Classification for Reliability Testing
MIL-STD-2076(AS)	Unit Under Test Compatibility with Automatic Test Equipment; General Requirements for
MS-17322-10	Meter, Time Totalizing, 115 Volt, 400 Cycle

PUBLICATIONS

Naval Air Systems Command

AVMP-184	ARBS Maintenance Plan
AV-093	ILSP for Angle Rate Bombing System (ARBS)

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2.1.1 Availability of Documents. When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors In connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

2.2 Non-Government Documents. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified herein, the issue in effect on the date of invitation for bids or request for proposal shall apply.

Hughes Aircraft Company
(Code Ident 82577)

DRAWINGS

3366OOO-310	ARBS/A-4M Aircraft Wire List
3366000-350	ARBS Electrical Interface Data
3366000-500	Installation Control - Digital Computer Unit, Revision D
3366011-500	Installation Control - Computer Interface Unit

International Business Machines Corporation
(Code Ident 03640)

SPECIFICATIONS .

6009484	Specification for Double Sides and Multilayer, Plated Through Hole, Printed Wiring Boards, May 1973

2.2.1 Availability of Non-Government Documents. Non-Government documents may be obtained from the following sources:

(1) Technical society and technical association specifications are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

(2) Requests for copies of Hughes Aircraft Company and International Business Machines Corporation documents should be addressed to the contracting officer.

3. REQUIREMENTS

3.1 Preproduction. This specification makes provision for preproduction testing.

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3.2 Parts and Materials. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Microelectronic technology shall be considered and microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirements shall conform to Specification MIL-E-5400.

(3) Nonrepairable subassemblies shall be used in accordance with Specification AR-10 and as outlined in Specification MIL-E-5400.

(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

(5) Transistors and diodes shall be chosen and applied l s outlined in MIL-STD-454, Requirement 30, except JAN TX grade shall be used.

3.2.1 Nonstandard Parts and Materials Approval . Approval for the use of nonstandard parts and materials (including electron tubes, transistors and diodes) other than microelectronic devices shall be obtained as outlined in Specification MIL-E-5400.

3.2.2 Microelectronic Modular Assemblies. When used, Microelectronic Modular Assemblies shall meet the requirements of Specification AR-5. Conformal coatings, encapsulant, embedments or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly.

3.3 Design and Construction. The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight end Dimension. The total weight of the-equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 46.5 pounds. The dimensions of the digital computer (DC) and computer converter-interface unit (CIU) shall be as specified on the following drawings:

(1) DC: 3366010-500

(2) CIU: 3366011-500

3.3.2 Reliability. The contractor shall conduct a reliability program using MIL-STD-785.

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3.3.2.1 Operational Stability. The equipment shall operate with satisfactory performance, continuously or intermittently for a period of 1800 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life. The equipment shall have a total operating life of 10,000 hours with reasonable servicing and replacement of parts.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF). The equipment Including any Built-In-Test provisions, shall have a specIfied MTBF of 1800 hours when tested and accepted as outlined under the requirements of 4.3.2. The minimum acceptable MTBF shall be 900 hours.

3.3.2.4 Calendar Life Limit.. The contractor shall identify all items that change significantly with age (have a calendar life limit that would fall within a 10-year period) and whose change would degrade the reliability of the equipment. This effort shall be coordinated with the limited life/age environment control requirement as specified in the contractor's quality program.

3.3.2.5 Failure Classification. Classification of failures shall be in accordance with MIL-STD-781 and MIL-STD-2074.

3.3.2.5.1 Failure Definition. A failure is defined as follows:

a. Failure to provide the performance specified in 3.4.

b. Any unplanned maintenance, adjustment, or alignment which is inaccessible to the operator.

c. Excessive wear or damage of mechanical components in accordance with the contract requirements.

3.3.2.5.2 Failure Reporting. The contractor's failure reporting, analysis and corrective action system shall apply to failures l xperienced on each subassembly and end-item during tests performed in accordance with approved test procedures. Failure analysis and corrective actions shall be included in the contractor's failure reports and shall be submitted to the procuring activity in accordance with the contract requirements.

3.3.2.5.3 Failure Analysis. The contractor shall perform failure analysis on all failures reported to the procuring activity to the in-depth level required to define the failure mechanism and to develop corrective actions to limit their recurrence. In general, the analysis shall include a physics of failure analysis on electrical parts. The analysis of parts shall typically include electrical failure verification, dissection, microphotography, and adequate chemical and metallurgical analysis to define the failure mechanism (e.g., most fundamental cause; metallization corrosion). Records of failure analysis, including causes and effects,

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shall be maintained by the contractor with data feedback to reliability and related design analysis functions.

3.3.2.5.4 Corrective Action. The contractor, in conjunction with failure analysis, shall develop and implement effective corrective action to limit recurrence of the failure mechanism and its effects. Corrective action for those failures reported must be approved by the procuring activity. Approval of proposed contractor corrective actions requires that the contractor has performed the following:

a. Analytically or by test established the effectiveness of the corrective action to the satisfaction of the procuring activity.

b. A schedule for the introduction of the above corrective action.

No individual deliverable articles for which failures are detected in acceptance tests shall be released for delivery to the procuring activity until the failure reporting and corrective action process has been initiated.

3.3.3 Cabling and Connections.

3.3.3.1 Cables and Connectors. The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.3.3.2 Interconnection Cabling. The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088.

3.3.4 Interchangeability. The equipment shall meet the interchangeability requirements of Specifications MIL-E-5400 and MIL-I-8500.

3.3.5 Interference Control. The "generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall meet the requirements of MIL-STD-461, Notice 3.

3.3.6 Maintainability Provisions. The maintenance concept for support of the equipment shall be compatible with the existing ARBS Maintenance Plan, AVMP-184. Maintainability features, built-in-test features, construction, provisions for test points, and other maintainability parameters specified herein, shall conform to AR-10, with the stated exceptions and clarifications specified herein. A maintainability program in accordance with MIL-STD-470 shall be established to enforce and report on the maintainability requirements of this specification.

3.3.6.1 Fault Diagnosis. Fault diagnosis at the organizational maintenance level shall be accomplished using Built-In-Tests (BIT) and displays. Intermediate level fault diagnosis and BIT shall be accomplished using ground support equipment designated, designed, and planned for Fleet use, and any combination thereof.

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3.3.6.2 Test Points. Test points shall be provided on the equipment for intermediate and depot levels of maintenance. The test points shall be compatible with the Ground Support Equipment (GSE) designated and planned for Fleet use. To provide for those situations where the GSE is not available, all hardware shall have sufficient exposed or accessible test points to permit use of standard Navy test equipment (oscilloscope, VOM, VTVM, etc.), to permit, in the event of weapon replaceable assembly (WRA)/shop replaceable assembly (SRA) substandard performance, ascertaining to the maximum extent possible without ambiguity which module, component, or group of components is at fault. Test points shall be exposed or accessible at each level of maintenance of the WRA as necessary, and sufficient for intermediate and depot level maintenance. The following guidelines shall also apply:

a. Duplication: Duplication of test points necessary for built-in-test equipment (BITE), calibration, alignment, adjustment, etc., is not required.

b. Protection: Each test point shall have loading or isolating characteristics incorporated to prevent circuit damage in the event any test point is accidentally connected to ground.

c. Safety: Safety design and precautions shall be in accordance with MIL-STD-454, Requirement 1, MIL-STD-882, and MIL-STD-1472.

d. Multiple test points: Where multiple tests are required on a WRA, multi-pin connectors may be used to facilitate connection of the GSE or other test equipment to the WRA.

3.3.6.3 Modularization. The equipment shall be modularized to provide WRAs and SRAs and be functionally modularized in accordance with AR-10.

3.3.6.3.1 Module Commonality. The number of different types of modules shall be kept to an absolute minimum. This standardization shall be mandatory to ease the testing and spares requirements at all levels of maintenance. Unused functions on a module are permitted if commonality is improved. The requirements for employing standard hardware program modules in accordance with MIL-STD-1378 shall be considered.

3.3.6.3.2 Module Accessibility. The access to each module shall not require the removal of adjacent modules or parts, other than access panels.

3.3.6.3.3 Module Pins. Where printed circuit card or modular construction is used, ground potential and operating voltages shall be assigned to pins of all plug-in assemblies in a uniform manner so that a given potential is applied to pins of the same number on all plug-in assemblies which utilize similar packaging.

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3.3.6.4 Adjustments and Alignments. All adjustments and alignments shall occur only during SRA repair. A list of all adjustments and alignments shall be submitted for approval by the procuring activity. There shall be no adjustments or alignments required for installation of WRAs into the system at the organizational level.

3.3.6.5 Maintainability Indices. The organizational and intermediate level maintainability indices shall conform to AR-10 with the stated exceptions of 3.3.6.7 and 3.3.6.4 and 3.3,6.12.

3.3.6.6 Reversibility. It shall be physically impossible to install any WRA or SRA or mate any cable Improperly. Mechanical keying, different size connectors and any other appropriate design implementation shall be used to minimize these possibilities.

3.3.6.7 Built-in Test (BIT) Function. Suitable BIT features shall be incorporated into the system or equipment to provide both a failure detection function and a failure location function.

3.3.6.7.1 Failure Detection Function. The failure detection function shall provide the equipment operator with a GO/NO-GO indication of equipment readiness.

3.3.6.7.2 Failure Location Function. In conjunction with, but not necessarily as a part of the Failure Detection Function, the equipment shall incorporate features which will automatically locate a failure both on the ground and in flight. This built-in failure location function shall be such as to obviate the need for any auxiliary test devices for the performance of organizational maintenance.

3.3.6.7.2.1 Failure Location Level. The failure location function shall locate the malfunction to the faulty WRA.

3.3.6.7.2.2 Fault Indicators. The failure location features shall automatically energize a GO/NO-GO signal device(s) located on the face of DC and CIU. The device(s) shall be such as to hold the last test result if power is interrupted or removed.

3.3.6.7.3 Built-in-Test-Equipment (BITE). The Weapon Delivery Computer (WDC) shall provide BITE hardware features that shall operate in conjunction with the ARBS Operational Flight Program (OFP)/Operational Test Program (OTP) to perform complete end-to-end checks, fault-isolation and performance monitoring of the WDC. The combined hardware/software test capability provided shall be such that at least 90 percent of the WDC failures shall be isolated to the appropriate WRA within 5 minutes, and 80 percent of the WDC failures shall be isolated to the appropriate WRA within 3 minutes. BITE hardware features shall

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be incorporated to provide GO/NO-GO Indications of equipment readiness automatically on the WRA front panel. These features shall be such as to energize a fault indicator (3.3.6.7.2.2) when equipment performance falls below a specified level. The DCU and CIU shall supply commands and other signals necessary to control and synchronize the operation of BITE functions in other subsystems, where appropriate. The circuits and devices which provide BITE functions shall be designed in such a manner that failure of the BITE circuitry shall not cause damage to any associated equipment or ground support equipment or cause unintentional operation of the associated equipment which would cause a hazardous condition.

3.3.6.8 Fail Safe Operation. The circuits and devices which provide BIT functions shall be designed in such a manner that to the extent possible, failure of the BIT circuitry will not cause a failure of the equipment with which it is associated.

3.3.6.9 Time-to-Repair. The m~tlme-to-repair (MTTR) a malfunctioned WRA at the organizational level to a specified performance level of operation shall not exceed 30 minutes with no more than 10 percent of the repair times exceeding 60 minutes. These MTTR specifications shall not, however, include failures due to combat damage and shall be independent of administrative and logistic delay times. The maximum MTTR shall not exceed 2 hours.

3.3.6.10 Automatic Test Equipment Compatibility. Maintenance provisions of the l quipment shall include compatibility with approved Fleet automatic test equipment. Requirements for special test equipment, tools and handling equipment shall be minimized. The use of Peculiar Ground Support Equipment (PGSE) to ensure ATE compatibility is permitted. The PGSE may include card extenders and adaptive circuits (Interconnecting devices) and may use existing test connectors.

3.3.6.11 Organizational (on-equipment) Maintenance. Organizational maintenance shall be performed on the equipment while installed in the aircraft and including all preventive and corrective maintenance in accordance with the ARBS Maintenance Plan, AVMP-184.

3.3.6.11.1 Memory Loader/Verifier (MLV). A memory loader/verifier will be used at the organizational level for loading/dumping the WDC operational flight program (OFP), and for verifying proper WDC operation. The MLV may also be used for loading special test programs for organizational level maintenance.

3.3.6.11.2 Scheduled Maintenance. Scheduled maintenance shall conform to the ARBS Maintenance Plan, AVMP-184.

3.3.6.11.3 Unscheduled Maintenance. Unscheduled maintenance consists of all corrective action resulting from observed system/component malfunctions or from defects discovered during scheduled maintenance/inspections.

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3.3.6.11.4 Organizational Maintenance Activities. Maintenance at the organizational level shall conform to the ARBS Maintenance Plan AVMP-184 and shall be limited to:

a. Fault identification by display indications during system Built-In Test/Operational Readiness Test (BIT/ORT) and in-place functional tests.

b. Isolation of the fault to one WRA, except for fault isolation ambiguities approved by the procuring activity.

co Correction or repair by removal and replacement of the defective WRA.

d. Verification of satisfactory post-repair operation.

e. Use of the Memory Loader Verifier to load/verify the Operational Flight Program.

3.3.6.12 Intermediate Level. Maintenance activities performed at the intermediate level shall be in accordance with the ARBS Maintenance Plan, AVMP-184. The MTTR shall be minimized through the use of easily replaceable modular assemblies. Repair of equipment shall include:

a. Verification of a fault to a defective WRA.

b. Isolation of the fault to a defective SRA.

c. Removal and replacement of defective SRAs.

d. Verification of satisfactory post-repair operation.

e. Repair of defective SRAs identified in AVMP-184.

f. Return WRA to service/supply.

3.3.6.13 Depot Level. Maintenance activities at the depot level shall be in accordance with AVMP-184.

3.3.6.14 Time Totalizing Meter. The equipment shall contain time totalizing meters in accordance with Specification MS-17323-10. The time totalizing meter shall be installed on the front face in a location directly visible without. requiring the removal of any connectors or opening the unit. The range of the meters shall be O to 9,999 hours, and they shall not be capable of being reset.

3.3.7 Nomenclature, Nameplates and Identification Marking. Nomenclature and serial number assignment, nameplate approval, and identification marking shall be in accordance with MIL-N-18307. Marking shall be in accordance with MIL-STD-130.

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3.3.7.1 Serlalization. Bach SRA shall have a permanent serial number. Serial numbers shall be applied at the final assembly of these units. In the event a unit is scrapped, the serial number shall not be reused.

3.3.8 Standard Conditions. The following conditions shall be used co establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room Ambient (25°C ± 5°C)
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Reference Signal	26 VAC + 2% - 5%, Phase C, 400 Hz
Input Power Voltage	115 VAC + 2% - 5%, 3Ø, 400 Hz 27.5 ± 0.5 VDc

3.3.9 Semite Conditions. The WDC, while operating or nonoperating, shall withstand the natural and induced environments as specified herein, Including l ll logical adverse combinations with repeated exposure to conditions having cumulative adverse effects.

3.3.9.1 Nonoperating Conditions. The WDC shall be capable of withstanding the temperature, humidity, vibration, and shock as shown in Table I without physical damage or subsequent degradation of performance.

3.3.9.2 Operating Temperature-Altitude. The WDC shall meet the performance requirements of 3.4 and 3.5 when exposed to the temperature-altitude environments shown in Figure 1, and Table 1, as follows:

(1) The WDC shall be capable of continuous operation at the extreme limits of Figure 1, Curve A. Continous operation is defined as indefinite operation within the extremes of Curve A.

(2) The WDC shall be capable of performing intermittent operation at the extreme limits of Figure 1, Curve B and at 80° C. Intermittent operation of the WDC is defined as 45 minutes (any selected mode) within the limits of Curve A, followed by 12 minutes operation (NAVIGATION mode) within limits of Curve B, followed by 6 minutes intermittent operation, in the WPN DEL mode, at sea level at 80° C, followed by 12 minutes operation (NAVIGATION mode) within limits of Curve B, followed by 45 minutes operation (any mode) within limits of Curve A.

(3) The WDC shall be capable of performing intermittent operations at the extreme limits of Figure 1, Curve C, without part failure, but performance degradation may occur. Intermittent operation of the WDC is defined as in (2), except that the limits of Curve C shall be substituted for the transient conditions of Curve B.

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TABLE I. Environmental criteria - factory to target.

Environment Item	Event - Condition									
	Transportation				Storage		Handling		Aircraft/Mission	
	Truck	Rail	Ship	Air	Airfield	Carrier	Airfield	Carrier	Non-operating	Op
Temperature High °C (°F) Low °C (°F)	50 (122) -30 (-22)	50 (122) -30 (-22)	38 (100) -18 (0)	60 (140) -18 (0)	70 (158) -30 (-22)	38 (100) 0 (32)	50 (122) -30 (-22)	30 (86) 0 (32)	85 (185) -54 (-65)	Figu -40 (
Shock °C/min	NEG	NEG	NEG	NEG	NEG	NEG	NEG	NEG	80	
Altitude-feet	NEG	NEG	NEG	50,000	NEG	S/L	NEG	S/L	50,000	S/L t
Air rate-ft/min	NEG	NEG	NEG	1,000	N/A	N/A	N/A	N/A	40,000	
Vibration Sinusoidal, g (PK) Freq. range (Hz)	2 g, 5-12 Hz 0.5 g, 12-100 Hz	1 g 5-500 Hz	0.7 g 5-70 Hz	0.4 g, 5-40 5 g, 40-700	N/A N/A	N/A N/A	N/A N/A	N/A N/A		
Vibration Random-g^2/Hz Freq. range (Hz)	See above				N/A	N/A	N/A	N/A	0.02 g^2/Hz (Figu	
Vibration Gunfire-g^2/Hz Freq. range (Hz)	N/A								(Figure 9) MIL-STD-810B, Meth RP: 0.02 g^2/Hz SDC: HE: 0.038 g^2/Hz CIU: DC: 0.04 g^2/Hz CU:	
Acceleration Shock (S) Load (L) Q (PK), time Waveform	Transportation (in container): 25 g, 25 ms, 1/2 sine wave Loading/unloading (in container): IAW MIL-STD-810B, Method 516, Procedure II								Shock 15 g 11 ms 1/2 sine wave	Lo Vert: Long: Lat:
Acoustic vib. dB									140 dB 53 - 6,784	
Humidity, lb water per lb dry air	100% humidity over a temperature range from 0°C (0.004 lb water per lb dry air) to 35°C (0.031 lb water per lb dry									

NEG = NEGLIBIBLE N/A = NOT APPLICABLE

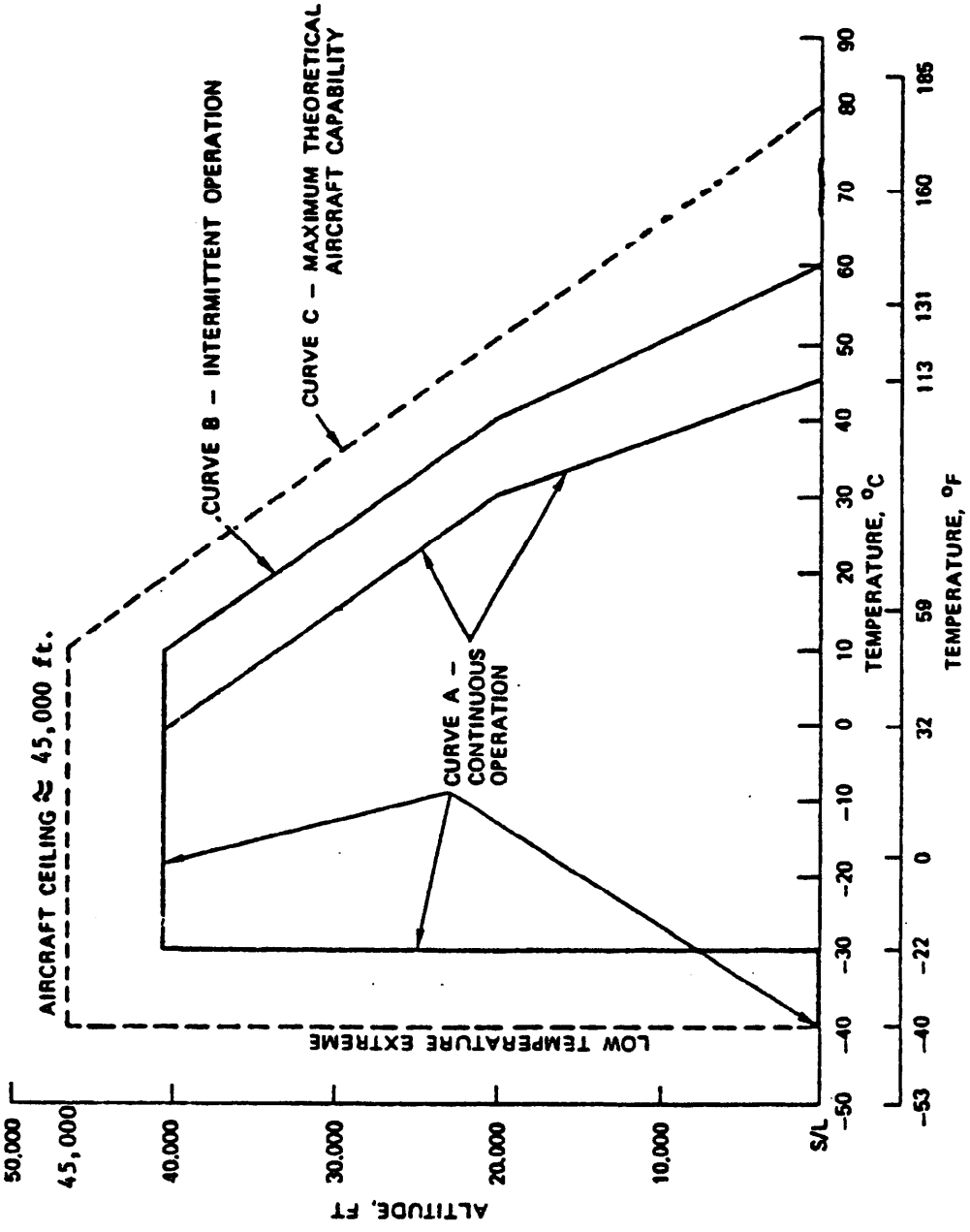


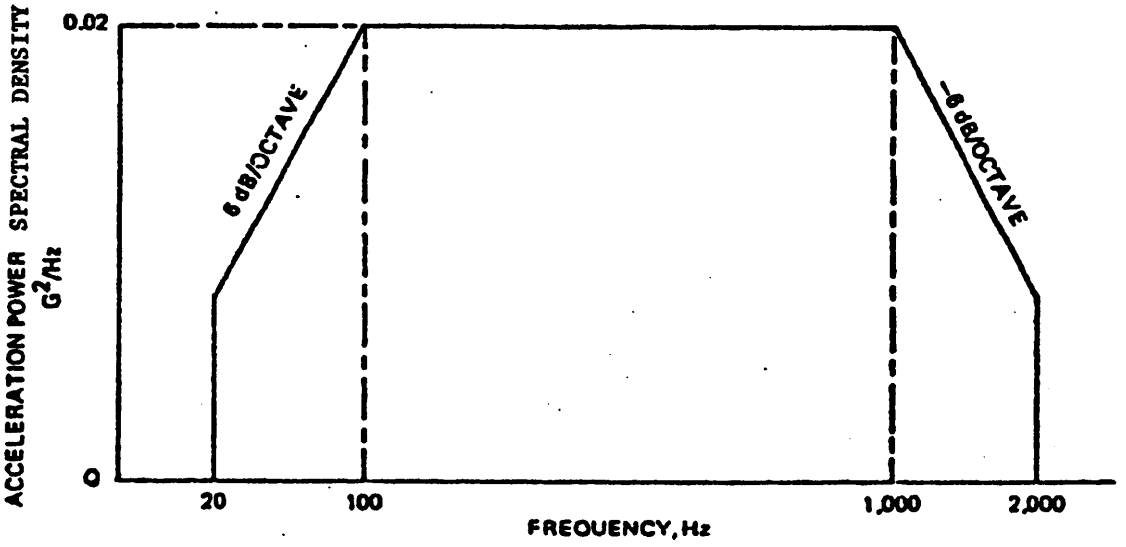
FIGURE 1. Equipment operating temperature/altitude.

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(4) The WDC shall be capable of starting at a low temperature condition of -40°C and perform as specified in 3.4 and 3.5 after the warmup period specified in 3.3.10.

3.3.9.3 Humidity. The WDC shall withstand the effects of humidity In accordance with Table I.

3.3.9.4 Vibration. The WDC shall not suffer damage and shall meet the performance requirements specified herein when subjected to random vibration specified in Figure 2.



OVERALL RMS G MINIMUM 5.3

FIGURE 2. Random vibration levels.

3.3.9.5 Shock. The WDC shall not suffer damage or subsequently fail to provide the performance specified herein when subjected to the shock specified in Table I.

3.3.9.6 Sand and Dust. The WDC shall withstand exposure to sand and dust as specified in MIL-E-5400.

3.3.9.7 Fungus. The WDC shall withstand exposure to fungus as specified in MIL-E-5400.

3.3.9.8 Salt Atmosphere. The WDC shall withstand exposure to salt-sea atmosphere in accordance with MIL-E-5400.

3.3.9.9 Explosive Condition. The WDC shall not cause ignition of an ambient-explosive-gaseous mixture with air when operating in such an atmosphere.

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3.3.9.10 Equipment Survivability (Crash Safety) - The surivability of the required systems shall be compatible, in all respects, with the A-4M aircraft and meet the requirements of MIL-E-5400.

3.3.9.11 Safety - Provisions shall be made to l nsure safety for personnel and equipment during all phases of operation or service use, storage, transportation, handling, test operations, and checkout. MIL-STD-454, Requirement 1, guidelines, MIL-STD-882, requirements, shall be used to ensure maximum safety through the production and operational use of the system. Safety hazard analysis and design considerations shall be In accordance with an approved Safety Plan, as prepared by the contractor, when specified in the contract. Fall-safe features shall be incorporated where failures nay disable the system, cause injury to personnel, damage equipment or cause inadvertent weapon release when installed in the aircraft. Protective devices shall be used on fans and blowers exposed to personnel during normal operation and maintenance.

3.3.9.12 Human Performance/Human Engineering - Human engineering design requirements shall be in accordance with MIL-STD-1472, except as approved by the procuring activity. System human engineering activities conforming to MIL-R-46855 shall be Implemented.

3.3.9.13 Acoustic Noise. The WDC shall be capable of performing in accordance with 3.4 and 3.5, after being subjected to sound pressure levels as specified in Table I, for periods as long as 30 minutes and shall suffer no physical damage as a result of exposure to acoustic wise.

3.3.9.14 Gunfire. The WDC shall be capable of withstanding, in the operating condition vibrations induced by gunfire as defined in Table I with no physical damage or degradation of performance.

3.3.10 Warmup Time. The time required for the WDC to warmup prior to operation shall be kept to a minimum and shall not exceed 30 seconds under the temperature conditions of Curves A and B of Figure 1, arid 5 minutes under conditions of Curve C of Figure 1.

3.3.11 Input Electrical Power.

3.3.11.1 Operating Power. The equipment shall give specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704. The 115/220-Vac power shall be switched on or off by a contactor in the CIU and controlled by the “WDC ON” signal. The power required shall not exceed the specified amounts.

(1) AC power (single Phase), not applicable (NA).

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(2) AC Power (Three Phase), Y-connected, 115/200 Volts (V), Category B, 500 volt amperes (VA).

(3) DC Power, 28 V, Category B, 25 Watts.

3.3.11.2 Degraded Performance. The line transient limits shall be as specified in MIL-STD-704, Figure 3, limit Curves 2 and 3, which defines the limits of transients within which the WDC maintains normal operation. Performance shall not be affected. by transients falling below 58 V whose duration is less than 75 microseconds (µs). After prime power remains at a level lower than 58 V for a period longer than 75µs, the WDC may go through a power OFF sequence. WDC operation shall return to normal when the voltage transient expires. No degradation to memory contents shall occur due to a power OFF/ON sequence.

3.3.11.3 Undervoltage Protection. The DC and CIU shall not be damaged by voltages below the minimum specified In MIL-STD-704 and shall automatically resume normal operation when the voltage returns within limits.

3.3.11.4 Power Failure Protection. Power monitoring and auto-restart circuits shall be provided. These circuits shall provide DC memory content protection in the event of power failure by delaying the DC shutdown until the present memory cycle has been completed. When power returns to normal, the DC shall be capable of automatically returning to a software initialization routine.

3.3.12 Cooling. WDC cooling, when required, shall be provided by a fan attached to each unit. The WDC shall require no special cooling other than ambient air of the normal installation environment.

3.3.13 Electrical Interfaces. The WDC electrical signal interface shall be as defined In Drawing 3366000-350. A summary of this interface definition is shown in Figure 3.

3.3.13.1 interface Connector Electrical Wiring. Shall be in accordance with Drawing 336600-310.

3.3.13.2 Aircraft Armament Signs1 Isolation. The following aircraft armament signals shall be electrically isolated from other operating and BITE circuitry to prevent inadvertent weapon release, in accordance with 3366000-350:

Master ARM	Fire Sequence
Bomb Button	Pullup Tone
Bomb Enable	Gun-pod Enable
RKT Enable	MSL Enable

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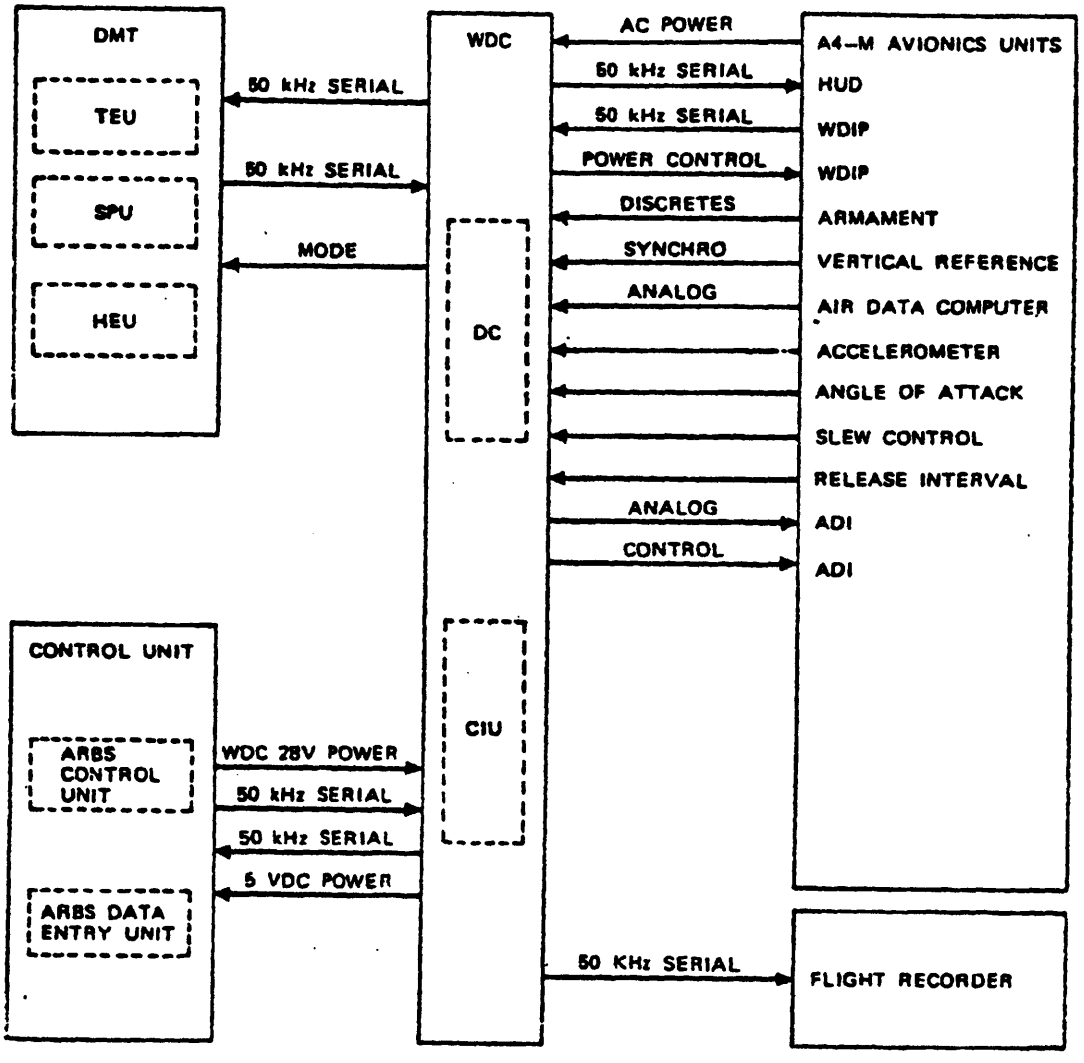


FIGURE 3. Summary definition of WDC interface.

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3.4 Performance. Unless otherwise specified, values set forth to establish the requirements of satisfactory performance apply to performance under standard conditions of 3.3.8, service conditions of 3.3.9, and Input power conditions of 3.3.11. When reduced performance under extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under standard conditions are specified.

3.4.1 Primary Performance Characteristics. The WDC shall provide primary performance characteristics for the following major functions;

(1) Memory.

(2) Instruction repertoire.

(3) Input/Output (I/O).

(4) Throughput.

(5) External Interrupts.

(6) Programmability.

3.4.1.1 Memory. The WDC main memory shall be random access nonvolatile ferrite magnetic core storage.

3.4.1.1.1 Word Length. Word length in memory shall be 16 bits plus one parity bit. The capability to operate on two words as a double word shall also be provided. An arithmetic data word shall be 15 bits plus a sign bit and the parity bit. An arithmetic data double word shall be 31 bits plus a sign bit plus 2 parity bits; one parity bit for each 16-bit word.

3.4.1.1.2 Parity Generation and Check. Odd parity shall be generated and stored for all data transferred to memory. Parity shall be checked for all data transferred from memory.

3.4.1.1.3 Memory Cross Temperature Read-Write Capability. The memory shall provide a cross temperature read-write capability for reading information at maximum temperature which has been written at minimum temperature and reading information at minimum temperature which has been written at maximum temperature. The maximum and minimum temperatures shall be determined by the temperature-altitude profile as specified in 3.3.9.2. This capability shall be independent of the status or past history of the memory.

3.4.1.1.4 Protection. The main memory shall be protected against any loss of contents due to power shutdown/activation cycles, power failure, or voltage transients as specified in 3.3.11.

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3.4.1.1.5 capacity and Expansion Provisions. The memory storage capacity provided shall be 16,384 words. The capability to expand the memory storage capacity to 32,768 words shall be provided wlthin the existing structure, within the existing power supply, logic and timing capability, and environmental requirements. The computer memory shall be fully tested for 32,768 word operation.

3.4.1.2 Instruction Repertoire. The WDC shall be capable of executing an instruction repertoire consisting of 43 instructions as identified in Table II of the following types:

(1) Loading/Storing of Registers.

(2) Arithmetic Operations.

(3) Logical Operations.

(4) Shift Operations.

(5) Branch (jump) Operations.

(6) I/0 Operations.

Table III contains the complete DC Instruction set, lncluding all addressing codes.

3.4.1.3 Timing. The timing shown shall be for Instruction, fetch, operand fetch (if required), and Instruction execution. For Indirect Addressing, 1.0µs shall be added to the time given. For Indirect and Tally, 1.5µs shall be added. For Immediate Addressing, 0.5 µs shall be subtracted. For base register, 0.25µs shall be added. For Modify Base (if Base Ø), 0.25 us shall be added. For Shift Instructions, N/8 us shall be added for N bits of Right Shift when N is even and (N + 1)/8 us when N is odd. For Left Shift, N/4 µs shall be added. For Long Format (RS) Instructions, 1.0 µs shall be added. For Branch Indirect (relative to Sector 2), 0.5 µs shall be added. For Branch Indirect (relative to Base or XC), 0.75 µs shall be added. .

As. a minimum, the execution of these instruction types shall provide the performance specified in Table 21.

3.4.1.4 Input/Output. The WDC shall provide the 1/0 capability that is required to implement the electrical interface definition specified in 3.3.13. A summary of the type and quantity of I/0 that shall be provided is presented in Table IV. As a minimum, the I0 processing of these I/0 types shall provide the performance specified herein.

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TABLE II. Alphabetical Listing of DC OP Codes

Mnemonic	Hex	Instructions	Execution Speed (microseconds)
A	8Ø	Add	2.0
AD	AØ	Add Double	3.0
AST	C8	Add to Storage	2.5
B	4Ø	Branch Unconditional	2.0
BAL	5Ø	Branch and Link	3.5
BF	Ø33	BITE Fail	1.5
BN	4C	Branch on A Negative	2.0
BO	44	Branch Out	2.0
BZ	48	Branch on A Zero	2.0
CE	7Ø	Compare (Skip on Equal)	2.25
CH	68	Compare (Skip on High)	2.25
CL	78	Compare (Skip on Low)	2.25
D	6Ø	Divide	6.0
DIN	Ø38	Direct In	2.5
DOUT	Ø68	Direct Out	2.5
IN	Ø8	In Service	2.5
LA	9Ø	Load A	2.0
LAB	Ø32	Load A from B	1.5
LB	2Ø	Load B	2.0
LBA	Ø31	Load B from A	1.5
LD	BØ	Load Double	3.0
LQ	3Ø	Load Q	2.0
M	F8	Multiply	4.0
MB	18	Modify Base	2.25
N	EØ	AND	2.0
NST	CØ	AND to Storage	2.5
O	E8	OR	2.0
OUT	1Ø	Out Service	2.25
RI	Ø3Ø	Reset Interface	1.5
S	88	Subtract	2.0
SD	A8	Subtract Double	3.0
SLDL	Ø34	Shift Left Double Logical	1.5
SLL	Ø36	Shift Left Logical	1.5
SRA	Ø37	Shift Right Arithmetic	1.5
SRDA	Ø35	Shift Right Double Arithmetic	1.5
STA	98	Store A	2.0
STB	28	Store B	2.0
STD	B8	Store Double	3.0
STQ	38	Store Q	2.0
T	58	Tally	2.5
TIO	D8	Tally I/O	2.25
X	FØ	Exclusive OR	2.0
XST	DØ	Exclusive OR to Storage	2.5

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TABLE III. Numerical listing of DC OP codes.

OP Code/Instruction	Mnemonic	Operand Address
ØBØØ	RI	—
ØB1Ø	LBA	—
ØB2Ø	LAB	—
ØB3Ø	BF	—
ØB4X	SLDL	—
ØB5X	SRDA	—
ØB6X	SLL	—
ØB7X	SRA	—
ØB8X–ØBFX	DIN	—
ØC8X–ØCFX	DOUT	—
Ø8XY	IN	ØØ8Ø+ØØXY FOR X ≤ 7 ØØ8Ø+FFXY FOR X ≥ 8
Ø9XY	IN@	C(IOT+ØØXY)
ØBXY	IN@	C(ØØXY)
ØCXY	IN	IOT+ØØXY
ØDXY	IN@	C(IOT+ØØXY), Tally Pointer
ØFXY	IN@	C(ØØXY), Tally Pointer
1ØXY	OUT	ØØ8Ø+ØØXY FOR X ≤ 7 ØØ8Ø+FFXY FOR X ≥ 8
11XY	OUT@	C(IOT+ØØXY)
13XY	OUT@	C(ØØXY)
14XY	OUT	IOT+ØØXY
15XY	OUT@	C(IOT+ØØXY), Tally Pointer
17XY	OUT@	C(ØØXY), Tally Pointer
18XY	MB	C(IC)+ØØXY IF X ≤ 7 C(IC)+FFXY IF X ≥ 8
19XY	MB@	C(Ø2XY)
1AØØ TTTT	MB$	TTTT
1BXY	MB	ØØXY
1CXY	MB	C(B)+ØØXY
1DXY	MB@	C(Ø2XY), Tally Pointer
1EXY	MBI	Immediate Operand: FFXY FOR X ≥ 8 ØØXY FOR X ≤ 7
1FXY	MB	Ø1XY
2ØXY	LB	C(IC)+ØØXY FOR X ≤ 7 C(IC)+FFXY FOR X ≥ 8
21XY	LB@	C(Ø2XY)
22ØØ TTTT	LB$	TTTT
23XY	LB	ØØXY
24XY	LB	C(B)+ØØXY
25XY	LB@	C(Ø2XY), Tally Pointer

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TABLE III. Numerical listing of DC OP codes (continued).

OP Code/Instruction	Mnemonic	Operand Address
26XY	LBI	Immediate Operand: ØØXY FOR X < 7 FFXY FOR X ≥ 8
27XY	LB	ØlXY
28XY	STB	C(IC)+ØØXY FOR X < 7 C(IC)+FFXY FOR X ≥ 8
29XY	STB@	C(Ø2XY)
2AØØ TTTT	STB?	TTTT
2BXY	STB	ØØXY
2CXY	STB	C(B)+ØØXY
2DXY	STB@	C(Ø2XY), Tally Pointer
2FXY	STB	ØlXY
3ØXY	LQ	C(IC)+ØØXY FOR X < 7 C(IC)+FFXY FOR X ≥ 8
31XY	LQ@	C(Ø2XY)
32ØØ TTTT	LQ$	TTTT
33XY	LQ	ØØXY
34XY	LQ	C(B)+ØØXY
35XY	LQ@	C(Ø2XY), Tally Pointer
36XY	LQI	Immediate Operand: ØØXY FOR X < 7 FFXY FOR X ≥ 8
37XX	LQ	ØlXY
38XY	STQ	C(IC)+ØØXY FOR X < 7 C(IC)+FFXY FOR X ≥ 8
39XY	STQ@	C(Ø2XY)
3AØØ TTTT	STQ$	TTTT
3BXY	STQ	ØØXY
3CXY	STQ	C(B)+ØØXY
3DXY	STQ@	C(Ø2XY), Tally Pointer
3FXY	STQ	ØlXY
4ØXY	B@	C(C(B)+ØØXY)
41XY	B@	C(Ø2XY)
42XY	B	C(IC)+ØØXY FOR X < 7 C(IC)+FFXY FOR X ≥ 8
43XY	B@	C(C(IC)+ØØXY) FOR X < 7 C(C(IC)+FFXY) FOR X ≥ 8
44XY	BO@	C(C(B)+ØØXY)
45XY	BO@	C(Ø2XY)
46XY	BO	C(IC)+ØØXY FOR X < 7 C(IC)+FFXY FOR X ≥ 8

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TABLE III. Numerical listing of DC OP codes (continued).

OP Code/Instruction	Mnemonic	Operand Address
47XY	BO@	C(C(IC)+ØØXY) FOR X < 7 C(C(IC)+FFXY) FOR X ≥ 8
48XY	BZ@	C(C(B)+ØØXY)
49XY	BZ@	C(Ø2XY)
4AXY	BZ	C(IC)+ØØXY FOR X < 7 C(IC)+FFXY FOR X ≥ 8
4BXY	BZ@	C(C(IC)+ØØXY) FOR X < 7 C(C(IC)+FFXY) FOR X ≥ 8
4CXY	BN@	C(C(B)+ØØXY)
4DXY	BN@	C(Ø2XY)
4EXY	BN	C(IC)+ØØXY FOR X < 7 C(IC)+FFXY FOR X ≥ 8
4FXY	BN@	C(C(IC)+ØØXY) FOR X < 7 C(C(IC)+FFXY) FOR X ≥ 8
50XY	BAL	C(ØØ8Ø+ØØXY) (Internal Interrupt)
51XY	BAL	C(Ø2XY)
54XY	BAL	C(ICT+ØØXY) (External Interrupt)
58XY	T	C(IC)+ØØXY FOR X < 7 C(IC)+FFXY FOR X ≥ 8
59XY	T@	C(Ø2XY)
5AØØ TTTT	T$	TTTT
5BXY	T	ØØXY
5CXY	T	C(B)+ØØXY
5DXY	T@	C(Ø2XY), Tally Pointer
5FXY	T	ØlXY
6ØXY	D	C(IC)+ØØXY FOR X < 7 C(IC)+FFXY FOR X ≥ 8
61XY	D@	C(Ø2XY)
62ØØ TTTT	D$	TTTT
63XY	D	ØØXY
64XY	D	C(B)+ØØXY
65XY	D@	C(Ø2XY), Tally Pointer
66XY	DI	Immediate Operand: ØØXY FOR X < 7 FFXY FOR X ≥ 8
67XY	D	ØlXY
68XY	CH	C(IC)+ØØXY FOR X < 7 C(IC)+FFXY FOR X ≥ 8
69XY	CH@	C(Ø2XY)

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TABLE III. Numerical listing of DC OP codes (continued).

OP Code/Instruction	Mnemonic	Operand Address
6AØØ TTTT	CH$	TTTT
6BXY	CH	ØØXY
6CXY	CH	C(B)+ØØXY
6DXY	CH@	C(Ø2XY), Tally Pointer
6EXY	CHI	Immediate Operand: ØØXY FOR X ≤ 7 FFXY FOR X ≥ 8
6FXY	CH	ØlXY
7ØXY	CE	C(IC)+ØØXY FOR X ≤ 7 C(IC)+FFXY FOR X ≥ 8
71XY	CE@	C(Ø2XY)
72ØØ TTTT	CE$	TTTT
73XY	CE	ØØXY
74XY	CE	C(B)+ØØXY
75XY	CE@	C(Ø2XY), Tally Pointer
76XY	CEI	Immediate Operand: ØØXY FOR X ≤ 7 FFXY FOR X ≥ 8
77XY	CE	ØlXY
78XY	CL	C(IC)+ØØXY FOR X ≤ 7 C(IC)+FFXY FOR X ≥ 8
79XY	CL@	C(Ø2XY)
7AØØ TTTT	CL$	TTTT
7BXY	CL	ØØXY
7CXY	CL	C(B)+ØØXY
7DXY	CL@	C(Ø2XY), Tally Pointer
7EXY	CLI	Immediate Operand: ØØXY FOR X ≤ 7 FFXY FOR X ≥ 8
7FXY	CL	ØlXY
8ØXY	A	C(IC)+ØØXY FOR X ≤ 7 C(IC)+FFXY FOR X ≥ 8
81XY	A@	C(Ø2XY)
82ØØ TTTT	A$	TTTT
83XY	A	ØØXY
84XY	A	C(B)+ØØXY
85XY	A@	C(Ø2XY), Tally Pointer
86XY	AI	Immediate Operand: ØØXY FOR X ≤ 7 FFXY FOR X ≥ 8
87XY	A	ØlXY
88XY	S	C(IC)+ØØXY FOR X ≤ 7 C(IC)+FFXY FOR X ≥ 8
89XY	S@	C(Ø2XY)

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TABLE III. Numerical listing of DC OP codes (continued).

OP Code/Instruction	Mnemonic	Operand Address
8AØØ TTTT	S$	TTTT
8BXY	S	ØØXY
8CXY	S	C(B)+ØØXY
8DXY	S@	C(Ø2XY), Tally Pointer
8EXY	SI	Immediate Operand: ØØXY FOR X < 7 FFXY FOR X ≥ 8
8FXY	S	ØlXY
9ØXY	LA	C(IC)+ØØXY FOR X < 7 C(IC)+FFXY FOR X ≥ 8
91XY	LA@	C(Ø2XY)
92ØØ TTTT	LA$	TTTT
93XY	LA	ØØXY
94XY	LA	C(B)+ØØXY
95XY	LA@	C(Ø2XY), Tally Pointer
96XY	LAI	Immediate Operand: ØØXY FOR X < 7 FFXY FOR X ≥ 8
97XY	LA	ØlXY
98XY	STA	C(IC)+ØØXY FOR X < 7 C(IC)+FFXY FOR X ≥ 8
99XY	STA@	C(Ø2XY)
9AØØ TTTT	STA$	TTTT
9BXY	STA	ØØXY
9CXY	STA	C(B)+ØØXY
9DXY	STA@	C(Ø2XY), Tally Pointer
9FXY	STA	ØlXY
AØXY	AD	C(IC)+ØØXY FOR X < 7 C(IC)+FFXY FOR X ≥ 8
AlXY	AD@	C(Ø2XY)
A2ØØ TTTT	AD$	TTTT
A3XY	AD	ØØXY
A4XY	AD	C(B)+ØØXY
A5XY	AD@	C(Ø2XY), Tally Pointer
A7XY	AD	ØlXY
A8XY	SD	C(IC)+ØØXY FOR X < 7 C(IC)+FFXY FOR X ≥ 8
A9XY	SD@	C(Ø2XY)

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TABLE III. Numerical Listing of DC OP Codes (Continued)

OP Code/Instruction	Mnemonic	Operand Address
AAØØ TTTT	SD$	TTTT
ABXY	SD	00XY
ACXY	SD	C(B)+ØØXY
ADXY	SD@	C(Ø2XY), Tally Pointer
AFXY	SD	Ø1XY
BØXY	LDD	C(IC)+ØØXY FOR X < 7 C(IC)+FFXY FOR X ≥ 8
B1XY	LDD@	C(Ø2XY)
B2ØØ TTTT	LDD$	TTTT
B3XY	LDD	ØØXY
B4XY	LDD	C(B)+ØØXY
B5XY	LDD@	C(Ø2XY), Tally Pointer
B7XY	LDD	Ø1XY
B8XY	STD	C(IC)+ØØXY FOR X < 7 C(IC)+FFXY FOR X ≥ 8
B9XY	STD@	C(Ø2XY)
BAØØ TTTT	STD$	TTTT
BBXY	STD	ØØXY
BCXY	STD	C(B)+ØØXY
BDXY	STD@	C(Ø2XY), Tally Pointer
BFXY	STD	Ø1XY
CØXY	NST	C(IC)+ØØXY FOR X < 7 C(IC)+FFXY FOR X ≥ 8
C1XY	NST@	C(Ø2XY)
C2ØØ TTTT	NST$	TTTT
C3XY	NST	ØØXY
C4XY	NST	C(B)+ØØXY
C5XY	NST@	C(Ø2XY), Tally Pointer
C7XY	NST	Ø1XY
C8XY	AST	C(IC)+ØØXY FOR X < 7 C(IC)+FFXY FOR X ≥ 8
C9XY	AST@	C(Ø2XY)
CAØØ TTTT	AST$	TTTT
CBXY	AST	ØØXY
CCXY	AST	C(B)+ØØXY
CDXY	AST@	C(Ø2XY), Tally Pointer
CFXY	AST	Ø1XY
DØXY	XST	C(IC)+ØØXY FOR X < 7 C(IC)+FFXY FOR X ≥ 8

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TABLE III. Numerical Listing of DC OP Codes (Continued)

OP Code/Instruction	Mnemonic	Operand Address
D1XY	XST@	C(Ø2XY)
D2ØØ TTTT	XST$	TTTT
D3XY	XST	ØØXY
D4XY	XST	C(B)+ØØXY
D5XY	XST@	C(Ø2XY), Tally Pointer
D7XY	XST	Ø1XY
D8XY	TIO	ØØ8Ø+ØØXY FOR X < 7 ØØ8Ø+FFXY FOR X ≥ 8
D9XY	TIO@	C(IOT+ØØXY)
DBXY	TIO@	C(ØØXY)
DCXY	TIO	IOT+ØØXY
DDXY	TIO@	C(IOT+ØØXY), Tally Pointer
DFXY	TIO@	C(ØØXY), Tally Pointer
EØXY	N	C(IC)+ØØXY FOR X < 7 C(IC)+FFXY FOR X ≥ 8
E1XY	N@	C(Ø2XY)
E2ØØ TTTT	N$	TTTT
E3XY	N	ØØXY
E4XY	N	C(B)+ØØXY
E5XY	N@	C(Ø2XY), Tally Pointer
E6XY	NI	Immediate Operand: ØØXY FOR X < 7 FFXY FOR X ≥ 8
E7XY	N	Ø1XY
E8XY	O	C(IC)+ØØXY FOR X < 7 C(IC)+FFXY FOR X ≥ 8
E9XY	O@	C(Ø2XY)
EAØØ TTTT	O$	TTTT
EBXY	O	ØØXY
ECXY	O	C(B)+ØØXY
EDXY	O@	C(Ø2XY), Tally Pointer
EEXY	OI	Immediate Operand: ØØXY FOR X < 7 FFXY FOR X ≥ 8
EEØØ	NOP	No Operation
EFXY	O	Ø1XY
FØXY	X	C(IC)+ØØXY FOR X < 7 C(IC)+FFXY FOR X ≥ 8

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TABLE III. Numerical listing of DC OP codes (continued).

OP Code/Instruction	Mnemonic	Operand Address
F1XY	X@	C(Ø2XY)
F2ØØ TTTT	X$	TTTT
F3XY	X	ØØXY
F4XY	X	C(B)+ØØXY
F5XY	X@	C(Ø2XY), Tally Pointer
F6XY	XI	Immediate Operand: ØØXY FOR X 7 FFXY FOR X 8
F7XY	X	Ø1XY
F8XY	M	C(IC)+ØØXY FOR X 7 C(IC)+FFXY FOR X 8
F9XY	M@	C(Ø2XY)
FAØØ TTTT	M$	TTTT
FBXY	M	ØØXY
FCXY	M	C(B)+ØØXY
FDXY	M@	C(Ø2XY), Tally Pointer
FEXY	MI	Immediate Operand: ØØXY FOR X 7 FFXY FOR X 8
FFXY	M	Ø1XY

IOT = I/O Tag
C(X) reads "contents of X"

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TABLE IV. WDC I/0 Summary

I/0 Type	Total No. Provided	No. Assigned	No. Of spares
1. DISCRETE INPUTS			
a. 5 V differential	1	1	-
b. 28 VdC	17	14	3
c. Switch closure	10	10	-
d. open circuit/ground	2		2
2. DISCRETE OUTPUTS			
a. Relay closure	8	8	
b. 28 Vdc	U	9	2
3. SERIAL DATA INPUT CHANNELS	3	3	-
4. SERIAL DATA OUTPUT CHANNELS	4	4	
5. SYNCHRO INPUTS	3	3	-
6. ANALOG DC INPUTS	lo	8	2
7. ANALOG DC OUTPUTS	2	2	
8. EXTERNAL INTERRUPTS (Single-Ended Inputs)	3		3

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3.4.1.4.1 Serial Data Channels. The serial data 1/0 channels shall be allocated as follows:

(1) Input Channels

a. Dual Mode Tracker (DMT)
b. Control Unit Subsystem (CUS)
c. Weapons Data Insert Panel (WDIP).

(2) Output Channels

a. DMT
b. CUS
c. Head-Up Display (HUD)
d. Flight Recorder.

The Input channels from the CUS and the WDIP, and the output channels to the CUS and HUD do not require concurrent operation. All other serial data channels shall be capable of concurrent operation.. Each channel shall have a separate line driver or receiver. Bach channel shall have the requirements for all digital interfaces, as stated in MIL-A-85055.

3.4.1.4.2 Growth Capability. The WDC shall provide the following growth capability:

(1) One serial data output channel.

(2) One serial data input channel.

(3) Two relay closure discrete outputs.

The defined characteristics of these 1/0 types shall be the same as the characteristics defined for these types in Table IV. This growth capability shall be implemented by the addition of a growth page and by the removal of jumper wires only. Future modifications or additions to existing hardware shall not be required In order to provide this capability. All needed wiring to the growth page connector shall be provided.

3.4.1.4.3 Data Formatting The WDC shall perform the data formatting required to transmit and receive signal data with the characteristics defined in 3.3.13.

3.4.1.4.4 Processing Modes. The WDC shall provide the following two modes for processing 1/0.

3.4.l.4.4.l Buffered InPut/Output. The transfer of I/0 data between the CIU and DC main memory shall be performed concurrent with DC instruction execution. The enablement of this mode, the number of words to be transferred, and the data storage or access location(s) in main memory shall be under software control.

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3.4.1.4.4.2 Direct Input/Output. The transfer of I/0 data between the CIU and the DC accumulator register shall be performed under software control.

3.4.1.4.5 Data Transfer Rates. The data transfer rates for the two processing modes of 3.4.1.3.4 shall be at least as follows:

Mode	Data Transfer Rate
Buffered I/0	125K words/second
Direct I/0	200K words/second

The data transfer rate for the buffered I/0 mode shall be obtained under the condition that no concurrent DC instruction execution is being performed.

3.4.1.5 Throughput. The WDC shall be capable of executing the instruction set defined in 3.4.1.2 at the speed specified in 3.4.1.3.

3.4.1.5.1 Throughput with Buffered Input/Output. The WDc shall execute requests for external buffered I/0 with an overhead not greater than 10 us for the first word in a block and not more than 5 us for each word thereafter.

3.4.1.6 Programability. The WDC shall be capable of being programmed by making use of the instruction set defined in 3.4.1.2.

3.5 Detail Requirements.

3.5.1 Digital Computer. The Digital Computer(DC), Type CP-1276/ASB-19(V), as a component of the WDC, shall provide the following:

3.5.1.1 Machine Organization. Machine organization within the DC shall contain, as a minimum, the following: a binary, fixed point, two's complement, fractional number system; full parallel (16 bits) operation; 6 registers (A, Q, B, IC, O/D, and INTF); and direct, indirect, indexed (base relative), instruction counter (IC) relative, and Immediate addressing modes. The A, Q, and B registers shall be capable of being loaded from or to memory, and the IC register shall contain the absolute address of the next memory location to be fetched as as instruction at the time that the current instruction is beginning execution. The contents of the XC register may be changed by branch Instructions and interrupts.

3.5.1.2 Direct Memory Access (DMA) Capability. The DMA data transfer rate shall be at least 500,000 words per second. The DC shall be capable of a DMA data transfer rate of 250,000 words per second and concurrently executing 150,000 instructions per second, using an Instruction mix of 76% ADD (1/2 Indirect), 18% Branch (1/2 Taken), 5% MDY (1/2 Indirect), and l% Divide (1/2 Indirect).

3.5.1.3 Memory.

3.5.1.3.1 Addressability. All words of the main memory shall be contiguously addressed from 0000 to 7FFF hexadecimal.

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3.5.1.3.2 Memory Speed. The memory read/restore cycle time shall be no more than 1.00 us maximum with a 500-nanosecond (ns) maximum direct access time. The memory shall operate in three modes as follows:

Mode	Function
Store:	Storage of correct data in an addressed location.
Read/restore:	Obtain data from an addressed location In storage without altering the contents of the storage location.
Read/compute/write:	Read data from an addressed location, cause memory to pause while the data is being used by the processor to generate new data, and then store the new data in the previously addressed location.

3.5.1.4 Central Processor (CP). The internal logic of the CP shall utilize data bits that are internally generated or transferred, or both, in a parallel mode, and all 1/0 comunications shall be executed in a parallel mode. The internal logic shall be capable of generating and handling both positive and negative numbers with l qual ease and execution time. Programmer access to the internal logic for all required operations shall be by means of the instruction set defined in 3.4.1.2.

3.5.1.4.1 Loading/Storing of Registers. Loading/storing from or into memory locations shall be accomplished from any one of three registers (A, B, or Q). These registers are defined herein.

3.5.1.4.2 Arithmetic Operations. The DC shall be capable of executing the following two's complement arithmetic operations:

(1) Addition: Adding of two single word operands to generate a single word sum, and adding two double length operands to generate a double length sum.

(2) Subtraction: Subtracting two single word operands to generate a single word difference, and subtracting two double length operands to generate a double length difference.

(3) Multiplication: Multiplying a single word multiplicand (positive or negative) by a single word multiplier (positive or negative) to yield a double word true product.

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(4) Division: Dividing a double word length dividend (positive or negative) by a single word divisor (positive or negative) to yield a single word true quotient.

3.5.1.4.3 Logical Operations. The following minimum logical operations shall be executed:

(1) Logical OR: Perform a bit-by-bit logical OR of a single word length operand and the accumulator register to produce a logical sum in the accumulator register.

(2) Logical AND: Perform a bit-by-bit logical AND of a single word length operand and the accumulator register to produce a logical product in the accumulator register.

(3) Exclusive OR: Perform a bit-by-bit exclusive OR of a single word length operand and the accumulator register to produce a logical sum in the accumulator register.

3.5.1.4.4 Shift Operations. The DC shall execute the following shift operations:

(1) Logical left shift: Shall shift the register contents of the accumulator left (least significant bit (LSB) toward the most significant bit (MSB) by N bits where N Is an integer between 0 and 15. Zeros shall be entered into vacated low order bit positions. Bits shifted out of high order bit positions shall be discarded.

(2) Arithmetic right shift: Shall shift the register content of the accumulator right by N bits with the sign bit being extended N places to the right. Bits shifted out of low order bit positions shall be discarded.

(3) Logical left shift double: Shall shift the register contents of the accumulator and the quotient register to the left by N bits with bits shifted out of the high order bit positions of the quotient register being shifted into the low order bit positions of the accumulator register. Zeros shall be entered into the vacated low order bit positions of the quotient register.

(4) Arithmetic right shift double: Shall shift the register contents of the accumulator and quotient right by N bits with bits shifted out of the low order bit positions of the accumulator register being shifted into the high order bit positions of the quotient register and the sign bit of the accumulator register extended N places to the right. Bits shifted out of the low order bit positions of the quotient register shall be discarded.

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3.5.1.4.5 Branch (Jump) Operations. The DC shall provide for, as a minimum, branch operations (jump shell herein be referred to as branch) in program execution address with the following operation:

(1) Unconditional branch: Shall have an instruction which loads the instruction counter register with a designated memory address and be possible to specify the address of any memory location.

(2) Conditional branch: Shall test the accumulator register either for contents negative (alternately equal to zero) and modify the instruction counter according to the results of the test.

(3) Subroutine branch: Shall transfer control to a subroutine at any point while saving the l ddress of the instruction following the subroutine branch Instruction and shall nest subroutine cane to at least four levels.

(4) Increment and branch: Shall increment a register or memory location and cause a branch to take place when the contents reach zero.

3.5.1.4.6 Input/Output Operations. The DC shall have an instruction that causes a single word operand to be placed on or accepted on the DC I/0 bus. The source or destination of the operand shall be the accumulator register.

3.5.1.5 Registers. As a minimum, the DC shall have six registers. Each register shall contain at least 16 bits. When more than the minimum number of registers are provided, they may share the functions below.

3.5.1.5.1 Accumulator Register (A-REG). The A-REG shall be a register that can be loaded from, or into memory, and:

(1) Contains the result of single precision sold/subtract operation.

(2) Contains the result of logical and shift operations.

(3) Contains one parameter for add/subtract and logical operations.

{4) Used in multiply/divide operations.

3.5.1.5.2 Base Register (B-REG). The B-REG shall coatain the base address and be loaded from storage, placed in storage, and added to operands from storage. Contents of the B-REG may be moved to or from the A-REG.

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3.5.1.5.3 Quotient Register Q-REG) . The Q-REG shall be a register that can be loaded from, or into memory, and:

(1) Contains half the source and result of double precision add and subract.

(2) supplies one parameter for multiply or half the dividend for divide.

(3) Shall contain half the product.

(4) Contains low half of double word dividend and is linked with A-REG.

3.5.1.5.4 Instruction Counter Register (IC-REG). The IC-REG shall contain the absolute address of the next memory location to be fetched as an instruction at the time that the current instruction is beginning execution. The contents of this register may be changed with the branch class of instructions.

3.5.1.6 Modes of Addressing. There shall be four modes of addressing a memory reference: direct, indirect, Indexed, and relative.

3.5.1.6.1 Direct. A part of the instruction word or a second word following the instruction refers to an absolute memory location. The minimum amount of memory addressable in this manner shall be 512 words.

3.5.1.6.2 Indirect. A part of the instruction word (DISP) shall be added to 512 to the address of a main store location (DISP + 512). The main store location contains the absolute address of the actual operand.

3.5.1.6.3 Indexed. A part of the instruction word shall be added to the contents of the to generate the operand l ddress.

3.5.1.6.4 Relative. A part of the instruction word shall be l dded to the contents of the IC-REG to generate the operand l ddress.

3.5.1.7 Interrupts. The DC shall accept a minimum of 11 interrupts having at least four priority levels as described below. The CP may inhibit all interrupts except Power Fail/Restart, Program Fault, and BITE Fail.

3.5.1.7.1 Power Fail/Restart Interrupt. When source power falls outside acceptable limits specified in the power supply for a period greater than 75µs, an interrupt of the priority level "O" shall be initiated. When power Is reapplied, the computer shall begin operation at the memory location for a priority level "O" interrupt. This interrupt shall force a hardware implemented system reset. It shall be possible to force a level "O" interrupt without cycling power by means of the AGE connector.

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3.5.1.7.2 Program Fault Interrupt. The appearance of a memory parity error shall generate an interrupt of priority level "I", namely a "break-in" to memory location 7, as will an attempt to address a non-existent memory location. A priority level "1" Interrupt shall initiate a timed logic path to set the FAULT INDICATOR.

3.5.1.7.3 BITE Failure Interrupt. When a BITE failure instruction is executed, a priority level "1" interrupt shall be generated as specified herein. Execution of a Reset Interface (RI) instruction within 2 milli-seconds (me) after "break-in" to location 7 shall inhibit setting the FAULT INDICATOR.

3.5.1.7.4 External Interrupts. Interrupt levels "2" and "3" shall be generated by the CIU.

3.5.1.8 System Timing. The DC shall provide timing signals required to establish the 50-kilohertz (kHz) timing for CIU I/0 functions. These shall be derived from a precision crystal oscillator located within the DC. The oscillator shell be accurate to within ± 100 parts per million (ppm) over the service conditions of 3.3.9.

3.5.2 Converter-Interface Unit. The CIU shall perform as specified herein.

3.5.2.1 General. The CIU shall provide the following:

(1) Synchro-to-digital signal Converslon.

(2) Analog-to-digital (A/D) signal conversion.

(3) Digital-to-analog (D/A) signal conversion.

(4) Interrupts.

(5) channel status.

(6) 50-kHz serial I/0 channel.

(7) 5 Vdc power for ARBS Control-Indicator.

3.5.2.1.1 Synchro-to-Digital Signal Converter. The CIU shall accept three separate synchro signal inputs and convert each of these to digital signals proportional to sin Ø cos Ø under program control. Synchro conversion accuracy shall be such that all errors in deriving Ø shell not exceed ± 3 milliradians. This accuracy shall apply over the required operating temperature range, with maximum signal Input voltages of 11.8 v mot—mean-square (me) ± 10 percent and with an input frequency of 400 Hz ± 5 percent.

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3.5.2.1.2 Analog-to-Digital Signal Converter. The A/D converter under program control shall be capable of converting two input signals concurrently. These two signals shall be multiplexed from five pairs of external analog inputs (including two spare inputs). Five pairs of signa1s from within the WDC shall also be selectable from the multiplexer for BIT purposes. The input impedance shall be greater than 10 megohms and the input voltage range shall be ± 10 V. The output format shall be sign magnitude with 14 bits, consisting of one sign bit, one overrange bit, and twelve magnitude bits. The bit weights shall be 10 V (nominal) for the overrange bit, 5 V (nominal) for the MSB of the magnitude. For any positive input voltage (Vin), the A/D output shall be l ccurate to within 30 mV of the calculated value, 10 Vin per volt reference (Vref), where Vref is the measured value of either positive reference voltage specified herein. For any negative Vin, the A/D output shall be accurate to within 30 mV of the calculated value, -10 Vin/Vraf; where Vref is the measured value of the negative reference voltage specified herein.

3.5.2.1.3 Conversion Time. For all inputs except the synchro inputs, the conversion time for a pair of inputs shall not exceed 1.7 ma. For synchro inputs, the total conversion time for l sin Ø and cos Ø pair of inputs, including the time required to synchronize with the 400-Hz reference voltages shall not exceed 4.4 me. The hardware shall be capable of completing three pairs of synchro-to-digital conversions and five pairs of A/D conversions in less than 19.6 ma.

3.5.2.1.4 Digital-to-Analog Signal Converters - Two D/A converters shall be included in the CIU and shall be under program control. Each of these converters shall have the following characteristics:

(1) Input format: Two's complement, parallel.

(2) Resolution: 11 bits.

(3) Accuracy: ± 5%

(4) Output voltage: ± 2.5 V.

(5) Output load capability: 1000 ohms, minimum

(6) Conversion time: 1.0 M, maximum.

3.5.2.1.5 Reference Voltages. The CIU shall provide the following reference signals which shall be derived from the ± 10 A/D converter reference Voltage. The internal reference shall be 10 V ± 60 mV. The magnitude of the - 10 Vdc reference shall be within ± 20 mV of the + 10 Vdc reference No. 2.

(1) + 10 Vdc reference No. 1: 25 miliamperes (mA), maximum.

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(2) + 10 Vdc reference No. 2: 75 mA, maximum.

(3) - 10 Vdc reference: 75 mA, maximum.

3.5.2.1.5.2 Maximum Reference Voltage. The reference voltages applied to associated equipment shall not exceed 14 V under all conditions of no load or internal power supply regulator failure, except that the maximum reference voltage applied to the Normal Accelerometer (see 6.6) shall not exceed + 13 V.

3.5.2.2 Converter-Interface Uuit Processed Interrupts. Priorities for Interrupt levels "2" and "3" shall be established by the CIU and provided to the DC as an external Interrupt request which will cause a "break-in" to the main store location "3". A level "2" interrupt shall have hardware priority over level "3" interrupts only if both occur at the same instant or are pending when interrupts are enabled. The priority scheme shall be implemented with a programmable mask which enables/disables individual Interrupts as bits in the mask are set to logical "1" or "0". The mask shall be readable and loadable by the program. Once an interrupt request has been acknowledged by the DC, all interrupts, levels "2" and "3", will be inhibited and a l6-bit data tag sent to the DC identifying the interrupt level as shown below:

(1) Priority level "2" (highest external priority order in CIU).

a. ISW bits Ø - 3
b. Linkage (HEX) Ø21Ø
c. Maskable.

(2) Priority level "3".

a. ISW bits 4 - 7
b. Linkage (HEX) Ø212
c. Maskable.

Interrupt requests shall remain inhibited until a Branch Out (BO) unconditional instruction is executed. Interrupt requests shall also be inhibited for a period of 40 to 60 µ following a data output to load the mask, unless BO is executed during this period, in which case the time period will be terminated and Interrupts enabled.

3.5.2.2.1 Converter-Interface Unit Supplied Interrupt Status Work (ISW). Interrupt levels "2" and "3" shall be identified in the CIU by means of an ISW. The allocation of bits in the ISW shall be as given below. The ISW shall be readable and bits independently resettable by the program. Interrupts will be accepted by the CIU even when interrupt requests to the DC are Inhibited.

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ISW BIT	A-REG BIT	Identificatlon	Priority Level
0	8	PROGRAM COUNTER NUMBER 2 (12 BIT)	2
1	9	PROGRAM COUNTER NUMBER 1 (16 BIT)	2
2	10	Go/No-Go	2
3	11	SPARE	2
4	12	CONVERSION COMPLETE	3
5	13	BUFFERED I/0 COMPLETE	3
6	14	SPARE	3

3.5.2.2.2 Real Time Clock (PLC 2). Within the CIU, there shall be a real time clock that decrements with a frequency of 50 kHz and an accuracy of ± 1 count in 10,000 or better. The counter shall be configured to allow generation of an interrupt at the lapse of a period between 100 µs and 80 ma. The current value of the counter shall be readable under program control. The counter shall be disabled by a "reset interface" signal and shall remain so until loaded by program control. Upon reaching zero, the counter will generate a level "2" interrupt, roll over to its maximum value, and continue to decrement. The interrupt shall be inhibited under program control by masking the interrupt. It shall be possible to maintain l measure of real time within the computer of 100 ppm or less over a 2-hour period. .

3.5.2.2.3 Internal Timer (PLc 1). Within the CIU, there shall be provided a down counter that can be loaded under program control to any value from 20 µs to 1.3 seconds. When the counter lapses to zero, a level "2" interrupt shall be generated and than discontinue operation. The counter is reactivated by reloading. When reloadad, the counter shall count down with a frequency of 50 kHz and an accuracy of at least ± 1 count in 10,000. The interrupt shall be inhibited under program control by masking the Interrupt.

3.5.2.2.4 GO/NO-GO (Watchdog) Timer. Within the CIU, there shall be a readable timer that measures a time equivalent to 1.3 seconds with an accuracy of at least ± 1 count in 10,000. This timer shall run whenever the computer is powered and not in manual control. The timer shall be resettable under program control and, when the time period lapses, a level "2" interrupt shall be generated. The current value of the counter shall be readable under program control. Upon issuing the interrupt, the counter shall continue to increment. The Interrupt shall be inhibited under program control by masking the interrupt.

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3.5.2.2.5 Conversion and Buffered InPut/OutPut (BIO) Complete. The completion of a requested conversion (A/D, D/A, synchro-to-digital) or of requested BIO shall initiate a level "3" interrupt, either of which may be inhibited under program control by masking the appropriate control bit .

3.5.2.3 Channel Status. The CIU shall generate a channel status word . (CSW) such as will identify the status of all CIU 1/0 channels as well as for BITE. The channel status word shall be readable and independently resettable under program control. ARBS channels shall be assigned bits within the CSW as shown in Table V.

Table V. ARBS CSW Bit Assignments

CSW Bit	Channel	I/0 Tag(Hex)	In/Out "1/"	Device
1	1A	0000	out	Heads-Up Display (HUD)
2	lB	0001	out	ARBS Control-Indicator
3	2	0002	out	Flight Recorder
4	3	0003	out	Dual-Mode Tracker (DMT)
5	4	0004	In	Dual-Mode Tracker
6	5A	0005	In	ARBS Control-Indicator
7	5B	0006	In	HUD Control-Processor
8	6	0008	out	Growth Spare
9	7	0009	In	Growth Spare

NOTES : 1. With reference to the CIU.

3.5.2.4 50-kHz Serial Input/Output Charnel. The CIU shall provide 50-kHz serial 1/0 channels allocated as defined in 3.4.1. The characteristics of the 50-kHz serial I/0 channels shall be as defined in Sec 3.4.1.4.1.

3.5.2.4.1 Buffered I/0 Transfers. During the Buffered I/0 (BIO) service operation three words of data will be transferred between the DCC and CIU. The first work shall be the I/0 tag identifying the serial channel which is requesting service, as shown in Table IV.

3.5.2.4.2 Transfer to DC. For BIO data transfers to the DC the next two words shall be address and data received from the serial device. The address shall consist of an 8-bit address and 4 flag bits which are assigned as follows for the word transfer to the DC.

Bits 0 -3	Don't Care
Bit 4	Serial Address Parity Error (1 = Error)
Bit 5	Serial Data Parity Error (1 = Error)
Bit 6	Serial Data Validity Bit (Bit = 1)
Bit 7	Serial Address Bit 19 (Bit = 0)
Bit 8	Serial Address Bit 18
Bit 9	Serial Address Bit 17
Bit 10	Serial Address Bit 16

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Bit 11	Serial Address Bit 15
Bit 12	Serial Address Bit 14
Bit 13	Serial Address Bit 13
Bit 14	Serial Address Bit 12
Bit 15	Serial Address Bit 11

The Bit positions for the serially receivred data words in the 16 bit transfer to the DC shall be:

Bit 0	serial Data Bit 18
Bit 1	Serial Data Bit 17
Bit 2	Serial Data Bit 16
Bit 3	serial Data Bit 15
Bit 4	Serial Data Bit 14
Bit 5	serial Data Bit 13
Bit 6	serial Data Bit 12
Bit 7	Serial Data Bit 11
Bit 8	Serial Data Bit 10
Bit 9	Serial Data Bit 9
Bit 10	Seriel Data Bit 8
Bit 11	Serial Data Bit 7
Bit 12	Serial Data Bit 6
Bit 13	Serial Data Bit 5
Bit 14	Serial Date Bit 4
Bit 15	Serial Data Bit 3

3.5.2.4.3 Transfer from DC. BIO data transfers from the DC also shall have the two words following the I/0 tag as address and data. Both words are 16 bits and are transferred to the CIU with bit placement is follOWs.

(Word form Dcu)	Bit o	serial Bit 18
	Bit 1	Serial Bit 17
	Bit 2	Serial Bit 16
	Bit 3	Serial Bit 15
	Bit 4	Serial Bit 14
	Bit 5	Serial Bit 13
	Bit 6	Serial Bit 12
	Bit 7	Serial Bit 11
	Bit 8	Serial Bit 10
	Bit 9	Serial Bit 9
	Bit 10	Serial Bit 8
	Bit 11	Serial Bit 7
	Bit 12	Serial Bit 6
	Bit 13	Serial Bit s
	Bit 14	Serial Bit 4
	Bit 15	Serial Bit 3

During final servicing of a serial output channel zero count issued) only one word is transferred from the DCU to the CIU. It is not used by the CIU.

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3.5.2.4.4 Channel Servicing. To assure a continuous flow of serial output data the three serial outputs shall be staggered such that DATA READY for each of the 3 channels are a minimum of 120 microseconds apart. Servicing of an output channel shall begin 20-40 microseconds before DATA READY is sent. Servicing must be complete by the and of DATA READY time. Serial input channels shall have buffering and therefore may be serviced anytime during the 400 microseconds necessary to Input (or output) a 20 bit serial word.

3.5.2.5 5-Vdc Power for ARM Control-Indicator. The CIU shall provide a 5-Vdc, 1.5-ampere power source for the ARBS Data Entry Unit. Characteristics of this power source shall be as defined in 3366030-350.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified, the contractor is responsible for the performance of all test requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the tests set forth in the specification where such tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests. Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction (First Article) Tests (See 4.2)

(2) Acceptance Tests (See 4.3}

(3) System Performance Demonstration Tests (See 4.4)

(4) Special Tests (See 4.5)

4.2 Preproduction (First Article) Tests. Preproduction tests shall be conducted by the contractor on equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The Government Inspector and the procuring activity shall be advised when tests are to be conducted so that a government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Reproduction (First Article) Test Data. The contractor shall submit data collected in conducting these tests to the procuring activity for review.

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4.2.2 Scope of Tests. Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications and the contract. Preproduction tests shall include:

(1) Individual Tests (See 4.3.1)

(2) Environmental Tests (See 4.2.2.1, 4.6.1)

(3) Preproduction Reliability Assurance Tests (See 4.2.2.2)

(4) Electromagnetic Interference Tests (See 4.2.2.3)

(5) Maintainability Demonstration Tests (See 4.2.2.4)

4.2.2.1 Environmental Tests. Environmental tests shall be performed in accordance with the procedure requirements of 4.6.1, to ensure that the requirements of 3.3.9, 3.3.10, 3.3.11, and 3.3.12 are met. The Environmental test requirements are listed in Table VI. The test sequence shall be determined by the contractor, subject to approval by the procuring activity. Environmental test shall follow Qualification Phase Test.

4.2.2.1.1 Certification. When specified in the contract, the following requirements may be certified by the contractor by submission for certification approval of design data, test data, test reports, analysis and other supporting documentation required to demonstrate compliance with the applicable requirements:

Fungus 3.3.9.7

Salt Atmosphere 3.3.9.8

Explosive Atmosphere 3.3.9.9

Equipment Survivability/Crash Safety 3.3.9.10

Safety 3.3.9.11

Human Performance/Engineering 3.3.9.12

4.2.2.2 Preproduction Reliability Assurance Tests (RAT). Preproduction RAT of the WDC shall consist of a Qualification Phase Test specified in 4.3.2.1.

4.2.2.3 Electromagnetic Interference (EMI) Tests. EMI tests shall be performed in accordance with MIL-STD-461, Notice 3 and MIL-STD-462, Notice 2.

4.2.2.4 Maintainability Demonstration Tests. The I-Level and O-Level maintainability requirements specified by 3.3.4, 3.3.6, and MIL-A-85055

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shall be verified by a maintainability test. The test requiremnets and procedures shall be prepared by the contractor and submitted to the procuring activity for approved. The test procedures shall include procedures-end requirements for adjustments and alignments required at all levels of assembly.

TABLE VI. ENVIRONMENTAL TESTS REQUIREMENTS

REQUIREMENT PARAGRAPH	METHOD PARAGRAPH	TITLE
3.3.9.1, 3.3.9.2	4.6.1.1	Temperature Shock
3.3.9.1. 3.3.9.2	4.6.1.2	Temperature-Altitude
3.3.9.1, 3.3.9.4	4.6.1.3	Low Temperature Vibration
3.3.9.3	4.6.1.4	Humitdy
3 .3.9.4	4.6.1.5	Vibration
3.3.9.5	4.6.1.6	Shock
3.3.9.14	4.6.1.7	Gunfire Vibration
3.3.9.13	4.6.1.5	Acoustic Noise
3.3.9.6	4.6.1.9	Salt Spray
3.3.9.6	4.6.1.10	Sand and Dust
3.3.9.9	4.6.1.11	Explosive Atmosphere
3.5.9.7	4.6.1.12	Fungus
3.3.15	4.6.1.13	Warm-Up Time
	4.6.2.4	Operating Power

4.2.3 Preproduction (First Article) Approval. Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample.

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4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as Imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Acceptance Tests - The contractor shall be responsible for accomplishing the acceptance tests. AU inspection and testing shall be under the supervision of the Government inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptmce tests. such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Reliability Assurance Tests

4.3.1 Individual Tests - Each WDC submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to detemine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a mimimum, each WDC accepted shall have passed the following tests:

(1) Examination of Product (see 4.3.1.1)

(2) Thermal Screening (See 4.3.1.2)

(3) Random Vibration (See 4.3.l.3)

(4) Operational Tests (see 4.3.1.4)

(5) Operational Mfr's Burn-in (See 4.2.1.5)

4.3.1.1 Examination of Product - Each WDC shall be examined carefully to determine that the material and workmanship requirements have been met. The tests shall consist of the physical and mechanical examinations listed in Table VII.

4.3.1.2 Thermal Screening - Each item covered by this specification shall be subjected to thermal screening tests as specified by MIL-A-85055.

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TABLE VII. Examination of Product requirements.

Requirement paragraph	Title	Preproduction acceptance	Production acceptance
MIL-A-85055			
3.2.4	Maintainabilility Provisions	X	
3.3	Design/Construction (MIL-E-5400)	X	
3.3.1	Parts and Materials (MIL-E-5400)	X	
3.3.1.2	Connectors	X	X
3.3.1.3	Adjustment Controls	X	
3.3.1.4	Printed Wiring Boards	X	X
3.3.1.5	Printed Wiring Assemblies	X	X
3.3.1.6	Elastomeric Materials	X	
3.3.1.7	Soldering	X	X
3.3.1.8	Wiring/Wire Coding	X	
3.3.1.9	Surface Finishes	X	X
3.3.1.10	Logic Circuits	X	
3.3.1.11	Screws and Fasteners	X	X
3.3.1 .12	Vibration Isolation	X	
3.3.1.13	Accessibility	X	
3.3.2.5	Bonding and Grounding	X	X
3.3.3	Nomenclature/Namelplates/ID	X	X
3.3.3.1	Serialization	X	X
3.3.4	Workmanship	X	X
3.3.4.1	Cleanup	X	X
3.3.5	interchageability	X	
3.3.8	Modules	X	
3.3.10	Cooling	X	
MIL-w-85057			
3.3.1.1	Total Weight	X	
3.3.1.1	Dimensions	X	X
3.3.6.15	Time Totalizlng Meter	X	X

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4.3.1.3 Random Vibration Tests - For the purpose of screening workmanship defects, each item covered by this specification shall be subjected to random vibration tests in accordance with MIL-A-85055.

4.3.1.4 Operational Tests - The WDC shall be subjected to the operational tests specified in Table VIII, in accordance with test procedures prepared and submitted by the contractor" and approved by the procuring activity. The test procedure requiremants are outlined in 4.6.2. During the operational tests, each unit under test shall be operated long enough to permit temperature stabilization and to provide sufficient time to check applicable characteristics and to record adequate data to assure satisfactory operation. Acceptance operational tests shall be performed prior to and after the completion of the Reliability Assurance (All Equipment Burn-in) tests.

4.3.1.5 Optional Manufacturer's Burn-in - Optional burn-in tests may be performed by the contractor in preparation for the reliability assurance tests.

4.3.2 Reliability Assurance Tests (RAT) - Reliability Assurance Test includes both Qualification and All Equipment Burn-In Test.

4.3.2.1Qualification Phase Test - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781B, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781B. For the qualification phase, test levels specified in paragraph 3.3.9 of this specification shall be used. The Accept-Reject Criteria for Test Plan IV shall be used. Failures shall be classified in accordance with MIL-STD-2074.

4.3.2.2 All Equipment Burn-In Test - Each equipment submitted for acceptance except those equipments submitted for qualifcation testing shall receive a minimum 200 hours burn-in with the last 100 hours failure free. Test level E of MIL-STD-781B shall be used except that random vibration of Figure 2 for 10 minutes shall be applied as a minimum in an axis or axes deemed most susceptible prior to and at the completion of the 100 hour failure free period. If a failure occurs during the failure free period the test shall be continued until 100 hours of test are failure free. All failures and failure free test time no matter how many repeats shall be totaled and the results compared with the reject line of Test Plan II of MIL-STD-781B. If the plotted data crosses the reject line, then the procuring activity shall be notified immediately and corrective action in accordance with paragraphs 5.4.8.5 and 5.7 of MIL-STD-781B shall be instituted. The test levels specified in paragraph 3.3.9 of this specification shall be used.

4.3.2.3 Test Incidents and Failure Reporting - All test incidents occuring after start of burn-in shall be investigated, failures analyzed, appropriate corrective action determined, and the following data provided:

a. Failure Reports

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TABLE VIII. Quality assurance performance (operational) tests.

Requirement paragraph	Method paragraph	Title	Preproduction tests	Acceptance tests
3.4.1.3.3	4.6.2.1	Memory Cross-Temperature Read-Write Capability	x	
3.3.16	4.6.2.2	Input Paver	x	
3.5.1.3.1	4.6.2.3	Addmissability	x	
3.3.11	4.6.2.4	Memory Protection Electrical Transients	x	
3.5.2.1.1	4.6.2.5	Synchro-to-Digital Conversion	x	x
3.5.2.1.2	4.6.2.6	A/D Conversion	x	x
	4.6.2.7	WDC Functional-Performance Tests	X	x
3.4.1.4	4.6.2.7	I/O	x	x
3.4.1.3.1	4.6.2.7	Serial Data Channels	x	x
3.4.1.3.2	4.6.2.7	Data Fomatting	x	x
3.4.2.4	4.6.2.7	Throughput	x	x
3.4.1.4.1	4.6.2.7	Throughput with BIO	x	x
3.5.1.9	4.6.2.7	system Timeing	x	x
3.5.2.1.1	4.6.2.7	sychro-tO-Digital Conversion	X	x
3.5.2.1.2	4.6.2.7	A/D Conversion	x	x
3.5.2.1.3	4.6.2.7	Conversion Time	x	x

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b. Failure Analysis Reports

c. Failure Analysis and Corrective Action Details and Summaries

d. Reliability Test Reports

Corrective actions shall be Implemented subject to approval by the cognizant Navy office. All failures shall be classified in accordance with MIL-STD-2074.

4.3.2.4 Failure Report - In the event of a failure the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review the failure reports and corrective actions and determine whether they are acceptable.

4.4 System Performance Demonstration Tests - Each WDC submitted for acceptance will be functionally tested, in accordance with MIL-A-85055, when integrated with the associated ARBS equipment. The functional tests will consist of demonstration of an approved functional test program or factory test program.

4.5 Special Tests - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control.

4.5.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.

4.5.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract.

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Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procurement action to require modification of such procedures, including additional tests, when deemed necessary.

4.6.1 Environmental Tests - The WDC shall be subjected to environmental tests in accordance with the environmental test plan and the procedures of MIL-STD-810, except as modified herein.

4.6.1.1 Temperature Shock - The requirements of 3.3.9.1 shall be verified by subjecting the WDC to temperature shock In accordance with MIL-STD-810, Method 503, and Table I.

4.6.1.2 Temperature-Altitude - The WDC shall be subjected to the temperature-altitude test in accordance with MIL-STD-810, Method 504, Procedure I for Class 2 equipment as modified by the requirements of 3.3.9.2. Intermittent operation shall be performed as follows:

(1) Raise chamber temperature to 60°C, stabilize equipment temperature, and operate for 12 minutes.

(2) Raise chamber temperature to 80°C and operate for 15 minutes. The WDC shall be operating during the transition to 80°C.

(3) Repeat test (1)

4.6.1.3 LOW Temperature Vibration - The requirement of Table I shall be verified at conditions of -40°C, 0.02 g-squared per Hz vibration level, and test time of 30 minutes.

4.6.1.4 Humidity - The WDC shall be tested in accordance with the test requirements of MXL-STD-810, Method 507, Procedure I, except as modified by the conditions of Table I, for a minimum 240 hours. The functional tests of 4.6.2.7.1 shall be conducted at the completion of the test.

4.6.1.5 Vibration - The WDC shall be subjected to 1 hour of random vibration, in accordance with MIL-STD-810, Method 514.1, and as specified in Figure 2, in each of three orthogonal axes. The WDC shall be operating during the vibration test.

4.6.1.6 Shock - The WDC shall be tested to the level specified in 3.3.9.5 in accordance with MIL-STD-810, Method 516.1, Procedure I.

4.6.1.7 Gunfire - The WDC shall be tested with simulated gunfire vibration in both an operating and nonoperating condition in accordance with the requirements of 3.3.9.11 and MIL-STD-810, Method 519, Procedure I. The functional test shall be performed during simulated gunfire vibration.

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4.6.1.8 Acoustic Noise - The WDC shall be subjected to acoustic noise in l ccordance with MIL-STD-810, Method 515. with the overall sound pressure level of Table I. The functional tests shall be conducted during this test.

4.6.1.9 Salt Atmosphere - The WDC shall be tested in accordance with MIL-STD-810, Method 509, Procedure I.

4.6.1.10 Sand and Dust - The WDC shall be subjected to sand and dust when nonoperating, in accordance with MIL-STD-810, Method 510, Procedure I. The functional test shall be performed prior to and after exposure to sand and duet.

4.6.1.11 Explosive Atmosphere - The WDC shall be tested in accordance with the test requirements of MIL-STD-810, Method 511, Procedure I. The functional test shall be performed during the exposure to the explosive environment . In lieu of testing, a certification that the WDC meets the explosion-proof requirement of 3.3.9,9 may be submitted.

4.6.1.12 Fungus Test - A certification that the requirements for fungus inert materials and components have been met may be submitted in lieu of performing a fungus test.

4.6.1.13 Warm-up Time - The preproduction WDC shall be tested to verify that the requirements of 3.3.10 are met.

4.6.1.14 Operating Power - The preproduction WDC shall be tested under worst case combinations of the input power conditions to verify compliance with the requirements of 3.3.11

4.6.2 Operational Tests - Each WDC shall be subjected to the operational tests specified herein, as listed in Table VIII.

4.6.2.1 Memory Cross Temperature Read-Write Capability - Memory data shall be stored at -40°C and read at +60°C and vice-versa to l ssure compliance with the requirements of 3.4.1.1.3. The WDC shall be stabilized at each temperature extreme prior to conducting the read-write operations.

4.6.2.2 Input Power - The total input powez to the WDC shall be measured and recorded at ambient temperature and at -40°C and +60°C.

4.6.2.3 Addressability - Contiguous addressability from 0000 to 3FFF hexadecimal shall be verified by 4.6.2.7. Contiguous addressability from 3FFF to 7FFF shall be verified.

4.6.2.4 Memory Protection/Eloctric~ Transients - The WDC shall be subjected to all extremes of electrical power and power transients as specified in 3.3.11.2 and 3.3.11.3 with the exception of transients with a duration of less than 75 us. Verification that the requirements of 3.4.1.1.4, 3.3.11.2, 3.3.11.3, and 3.3.11.4 are met shall be demonstrated through the execution of the computer test program specified in 4.6.2.7.

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4.6.2.5 Synchro-to-Digital Conversion - Three separate synchro signals proportional to sin Ø and cos Ø shall be simultanaously applied to the WDC. The WDC derived value of each Ø and the conversion time shall be measured by computer test program specified in 4.6.2.7.1 to varify the requirements of 3.5.2.1.1 and 3.5.2.1.3. The conversion accuracy shall also be verified In the same manner at extreme high (+60°C) l nd low temperatures (-40°C) during qualification test.

4.6.2.6 Analo&to-Digital Conversion -Analog input signals in the range from -10 Vdc to +10 Vdc shall be applied to the WDC. For each analog input signal, conversion accuracy and conversion time measurements shall be made by the computer test program specified in 4.6.2.7 to verify the requirements of 3.5.2.1.2 and 3.5.2.1.3.

4.6.2.7 Weapon Delivery Computer Functional/Performance Test - Functional/ performance tests shall be performed on each WDC to verify that the Section 3 requirements are met. A functional test program, operating in the DC, shall be used In conjunction with the common support equipment. The functional test program shall be lpproved by the procuring activity.

4.6.2.7.1 Functional/Perfornunca Veriftcation - The functional test program shall include a pass/fail history for Section 3 requirements defined by Table III to be tested (or group of requirements if tested collectively). Demonstration of the correct execution of the functional test program and examination of the pass/fail histories shall be used to verify that the Section 3 requirements are met. In addition, additional test data/measurements shall be obtained on the first unit undergoing functional performance testing to verify the functional test program algorithms used in testing the following requlrements:

(1) I/0 (3.4.1.4).

(2) Serial channels (3.4.1.4.1).

(3) Data formatting (3.4.1.4.3).

(4) Throughput (3.4.1.5).

(5) Throughput with BIO (3.4.1.5.1).

(6) System timing (3.5.1.8).

(7) Synchro-to-Digital signal converter (3.5.2.1.1).

(8) A/D signal converter (3.5.2.1.2).

(9) Conversion (3.5.2.1.3).

Verification that the requirements of 3.3.6.2.2 (BITE) are met shall only require the demonstration of the performance of the associated WDC hardware features.

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4.6.2.7.2 Timing- While operating the computear test program of 4.6.2.7, the 50-kHz clock outputs from the WDC shall be measured to ensure conformance to 3.501.8.

4.6.2.7.3 Reference Voltages - Measurements shall be made of the three reference voltages at full load to verify that the requirements of 3.5.2.1.5 are met.

4.6.2.7.4 5-Vdc Power Output - The WDC 5-Vdc output, specified in 3.5.2.5, shall be verified under no-load conditions. Each power supply shall be tested uncles full load for ± 2 percent tolerance on the 5-Vdc output during subassembly testing.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to initial production, acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

5. PREPARATION FOR DELIVERY

5.1 General. Unless otherwise specified in the contract or order all major units and parts of the equipment shall be preserved, packaged, packed and marked for Level A in accordance with Specifications MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794 and Method II A of MIL-P-116.

6. NOTES

6.1 Intended Use - The WDC is intended to be used to perform the computational functions for the AN/ASB-l9(V) Angle Bomb Bombing Set as used in the A-4M aircraft.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein, end procurement documents should specify the following:

(1) Title, number, and date of this specification.

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(2) Responsibility for inspection (see 4.1).

(3) Test data (see 4.5.4).

(4) Failure reports (see 4.5.4.1).

(5) Test procedures (see 4.6).

(6) Selection of applicable levels of packaging and packing (see 5.1).

6.3 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

a. Contract - The contract shall have precedence over any specification.

b. This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the cognizant Navy office.

c. Referenced Specifications. Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and an Improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts, and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Type Designations - The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.7. The correct type number shall be used on nameplates, shipping records, and Instruction books, as applicable.

6.6 Associated Equipment - The items listed herein are equipments with which the WDC is associated.

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ASSOCIATED EQUIPMENT FOR A-4M

Equipment	AN Designation or Procurement Spec
(1) Attitude Reference System (ARS):	AN/AJB-3A
(2) Remote Attitude Director Indicator (ADI):	ID-811/AJB-3A
(3) Relative Wind Transducer	DAC-7664058-5O3
(4) Weapon Release System . Programmer:	MX-7016/AWE-1
(5) Signal Data Converter	CV-3393/ASB-19 (V)
(6) Aircraft Throttle Handle/ Slew Controller	P/N DAC 5942590-1
(7) Aircraft Control Stick Grip	P/N DAC 5941954-2
(8) Normal Acclerometer	ABU-13/A
(9) Control Indicator (ARBS)	ID-2085/ASB-19(V)
(10) Air Data Computer (ADC)	AXC-666A-102
(11) Aircraft Armament systems	LAW DWG X3366000--350
(12) Head-Up Display (HUD):	AN/AVQ-24B
(13) Control Processor	C-XXXX/AVQ-24B
(14) Memory Loader Verifier (MLV)	AN/ASM-607

.6.7 Definitions of terms.

6.7.1 Unit. A removable box, made of modules, that connects with other units by external cabling and connectors, as typified by a gyro platform unit, a horizontal display unit, an interface unit, etc.

6.7.2 Modules. A subdivision of a unit characterized by the ease with which it can be removed by the application of a screwdriver to hold-down screws or clamps.

6.7.3 Card. An item removable by mechanical disconnect and made up of microelectronic Integrated circuits (MIC), components, and elements.

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6.7.4 Microelectronic Integrated Circuits (MIC). An integral silicon planar epitaxial structure that performs functions such as those of gates, amplifiers, adders, hi-stable elements, etc.

6.7.5 WRA. Weapons Line Replacement assembly; an assembly which for maintenance purposes is replaceable at the flight line or flight deck.

6.7.6 SRA. Shop Replaceable Assembly - a module for maintenance purposes is replaceable in the line shop or intermediate level of maintenance.

6.7.7 Failure. Cessation of equipment operation or degradatioa of a function beyond specified limits. Failure shall be considered either relevant or nonrelevant in accordance with the following criteria:

a. Nonrelevant failures as defined below shall not be classified as failures.

(1) Malfunctions caused by the operation of the equipment outside the specified electrical, mechanical, or temperature limits.

(2) Malfunctions caused by the operation or malfunction of equipment(s) external to the WRA.

(3) Malfunctions which are humanly induced through negligence, accidental overstress, Improper connection, damage in handling, installation or operation.

(4) Apparent malfunctions which result from programing errors.

(5) Malfunctions that have exhibited intermittent symptoms which cannot be verified or cannot be confirmed by reoccurrence within 24 hours of fault-free operation.

b. All failures not covered in (a) above shall be considered relevant.

6.7.8 C A T Customer Acceptance Test.

6.7.9 FAT - Factory Acceptance Test.

6.7.10 Abbreviations - The abbreviations and symbols used herein are defined as follows:

ac . . . . . alternating current

A/D . . . . . ,malog-to-digital

ADI . . . . . Altitude-Director Indicator

ARBs . . . . . Angle Rate Bombing System

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BITE . . . . . built-in-test equipment

CIU . . . . . Computer-Interface Unit

D/A . . . . . digital-to-enalog

dB . . . . . decibels

dc . . . . . direct currenmt

DC . . . . . Digital Computer

DMT . . . . . Mode Tracker

EMC . . . . . electromagnetic compatibility

EMI . . . . . electromagnetic interference

g . . . . . gravity unit

GSE . . . . . ground support equipment

HUD . . . . . Head-up Display

Hz . . . . . hertz

I/0 . . . . . input/output

k . . . . . kilo

LSB . . . . . least significant bit

MEA . . . . . Maintainence Engineering Analysis

MSB . . . . . most significant bit

MTBF . . . . . mean -time-between-failure

MTTR . . . . . mean-time-to-repair

NA . . . . . not applicable

PGSE . . . . . Peciliar Ground Support Equipment

PLC . . . . . program loadable Counter

PLc . . . . . program loadable counter 2

PPm . . . . . parts per million

RWT . . . . . Relative Wind Transducer

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SRA . . . .shop replaceable assembly

V . . . .VOlt

VAST . . . . . .Versatile Avionics Shop Test System

VOM . . . .volt ohmmeter

VTVM . . . . . vacmuum-tube voltmeter

WDc .. . .Weapon Delivery Computer

WRA . . . . . .weapon replaceable assembly

°c . . . .degrees Celsius

6.8 Revisions - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.9 Coganizance. This specification is under the cognizance of Naval Air Systems Command, Avionics Division (AIR-533), Washington, D. C. 20361.

Preparing Activity
(NAVY)

Project No. 128O-NO1O

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE: MIL-W-85057(AS) Weapons Delivery Computer for the Bombing Set, Angle Rate AN/ASB-19(V)

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED S/N 0102-014-1802

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